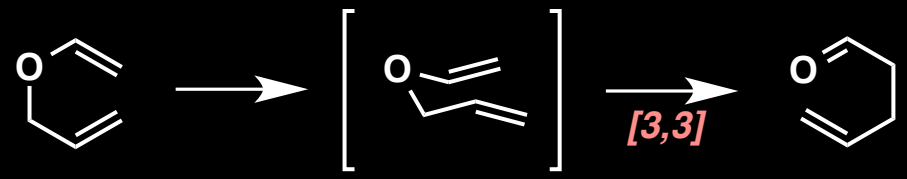
The Claisen Rearrangement

Lecture Notes



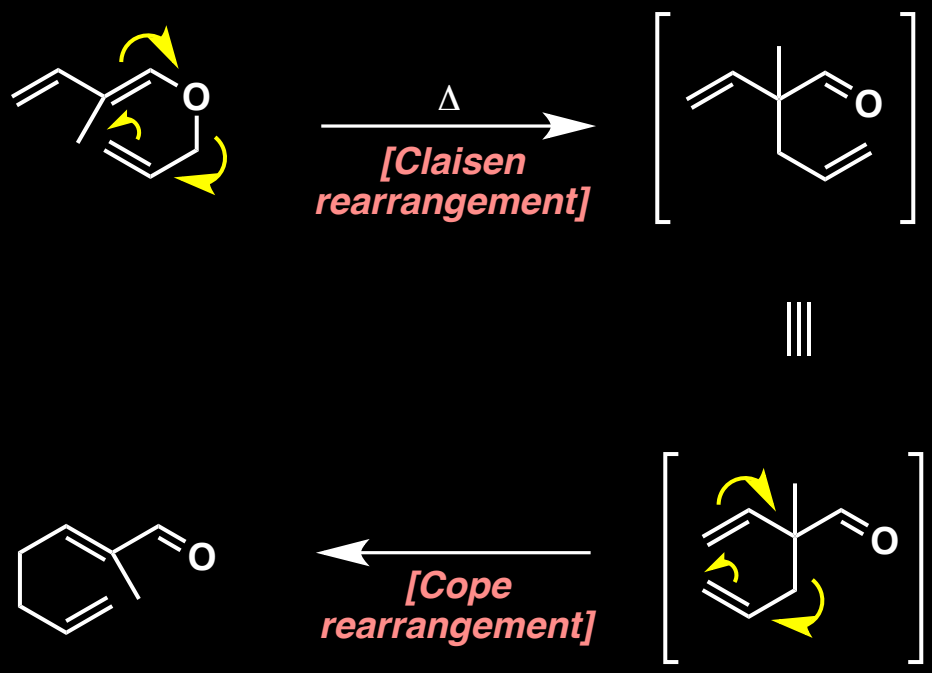
Key Reviews:

F. E. Zeigler, Chem. Rev. 1988, 88, 1423.

P. Wipf in Comprehensive Organic Synthesis, 1991, vol. 5, p. 827.

A. M. Castro, Chem. Rev. 2004, 104, 2939.

[3,3]-Rearrangements: The Claisen and Cope Reactions/Rearrangements



The Claisen Reaction: Initial Discovery



Bonds Broken	Bonds Formed
2 C=C	1 C=C
1 C–O	1 C=O
	1 C–C

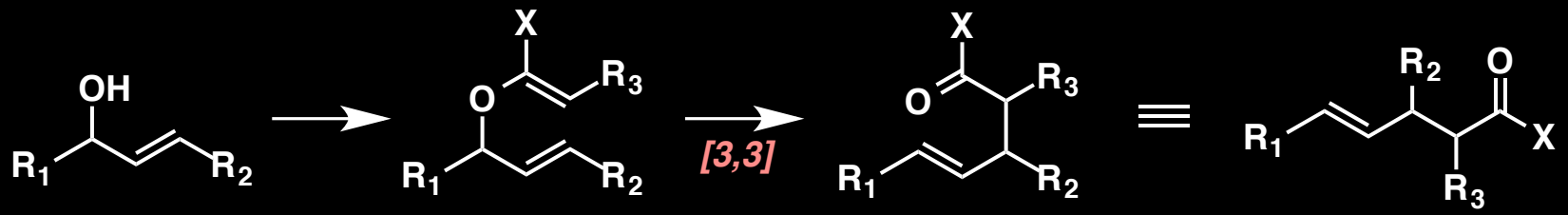
L. Claisen, Chem. Ber. 1912, 45, 3157.

The Claisen Reaction: Initial Discovery



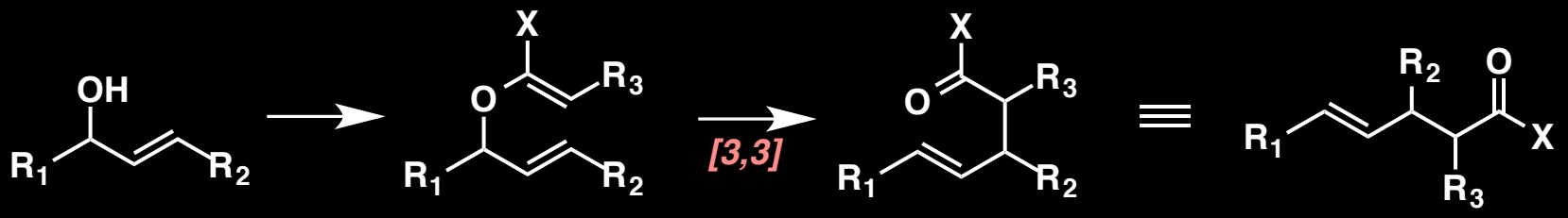
Bonds Broken	Bonds Formed
2 C=C	1 C=C
1 C–O	1 C=O
	1 C–C

Reaction can be regarded as the sigmatropic homologation of an allylic alcohol, meaning the migration of a sigma bond across a conjugated pi system to a new position.

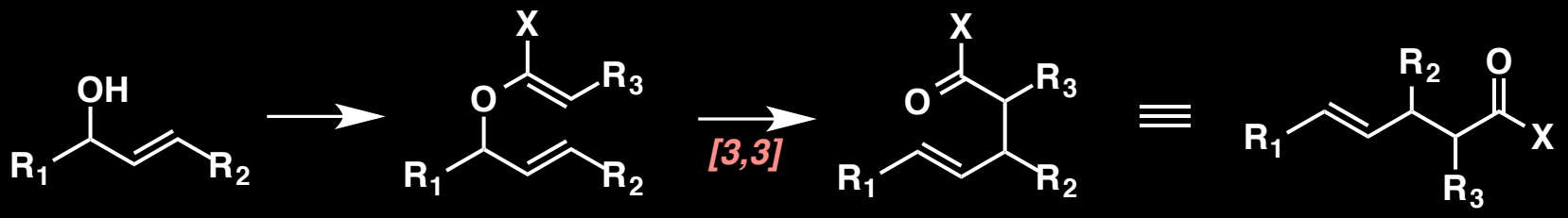


L. Claisen, Chem. Ber. 1912, 45, 3157.

The Claisen Reaction: Why is it So Useful?



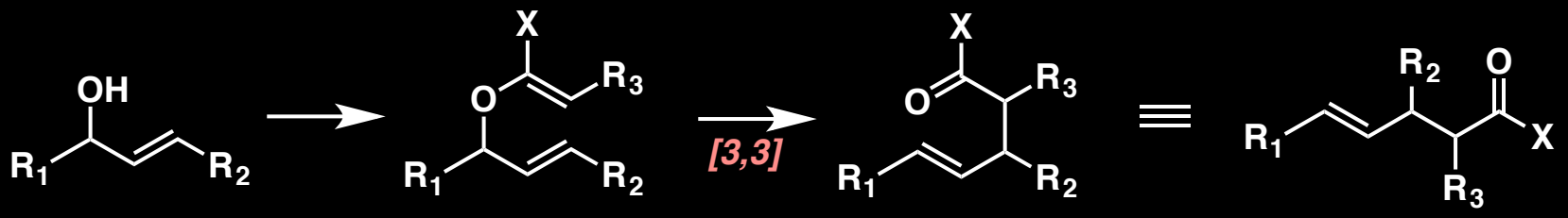
The Claisen Reaction: Why is it So Useful?



Key points:

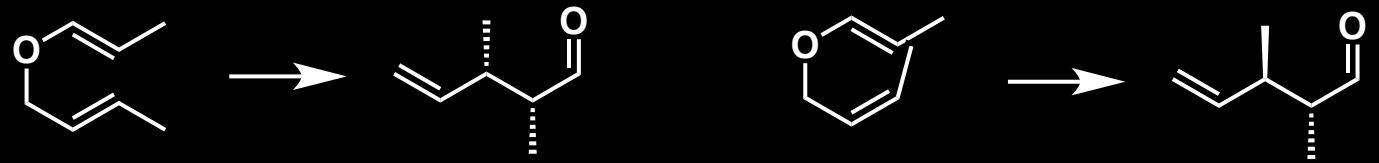
1. Experimental procedures are very reliable
2. Compatible with a wide range of functional groups
3. Substrates are readily made, and the δ,γ-unsaturated product can be chemoselectively adjusted at either end (alkene or carbonyl) to give further materials
4. High stereoselectivity in formation of double bonds and chiral centers.

The Claisen Reaction: Why is it So Useful?

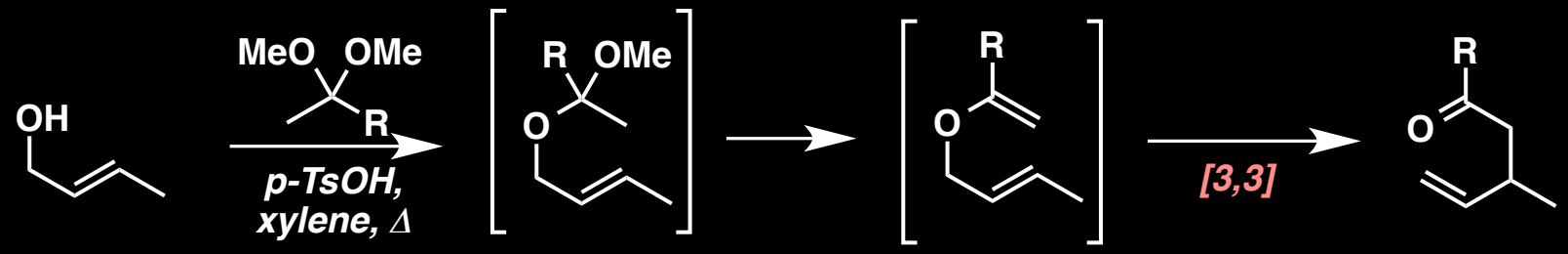


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The Claisen Reaction: Substrate Preparation



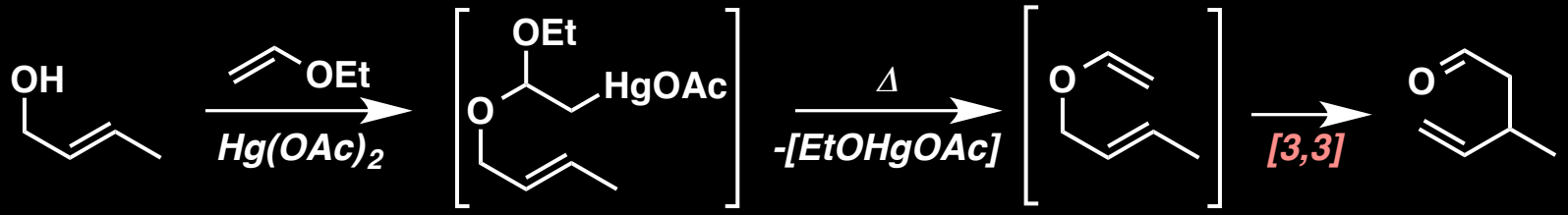
Potential drawbacks

Competing acid-catalyzed reactions with resident functional groups

If R is an alkyl group other than methyl, regiochemical problems in vinyl ether synthesis can occur

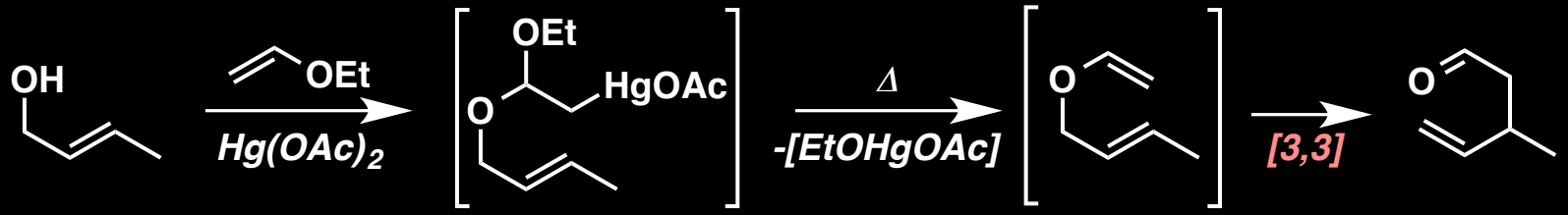
J. Org. Chem. 1976, 41, 3497; J. Org. Chem. 1976, 41, 3512; J. Org. Chem. 1978, 43, 3435.

The Claisen Reaction: Substrate Preparation

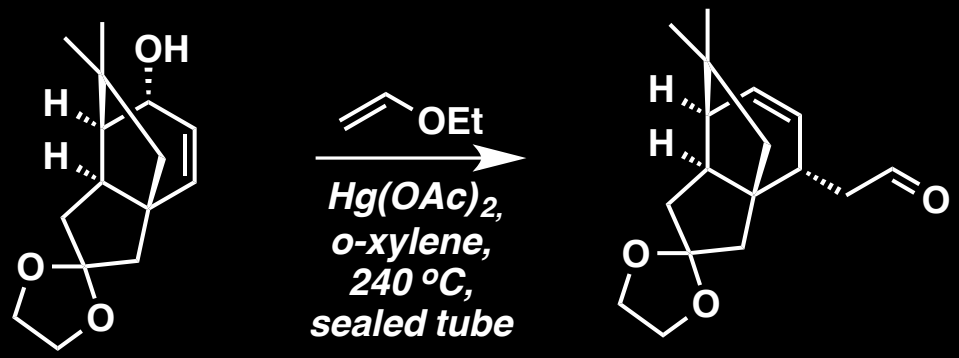


S. D. Burke and co-workers, Strategies and Tactics in Organic Synthesis 1987, 2, 57.

The Claisen Reaction: Substrate Preparation

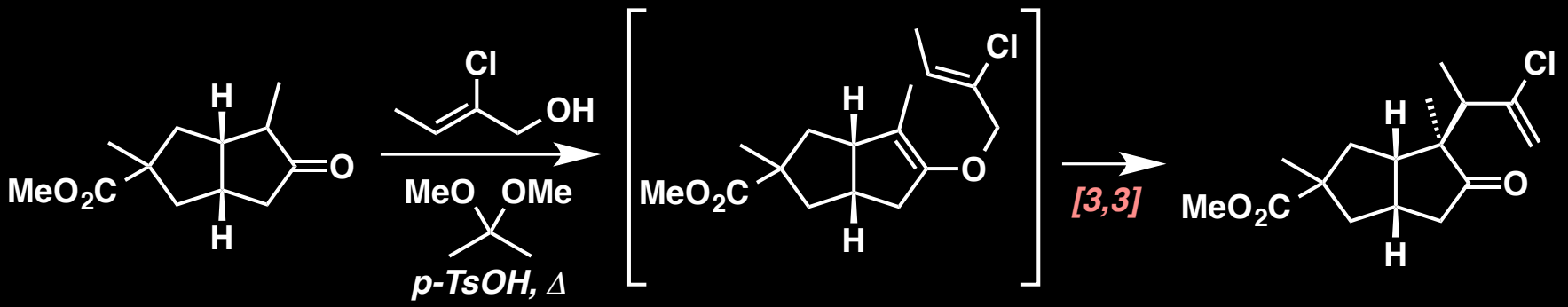


A much milder procedure for cases where substrates cannot tolerate acid and heat; in the case below it succeeded where others failed.



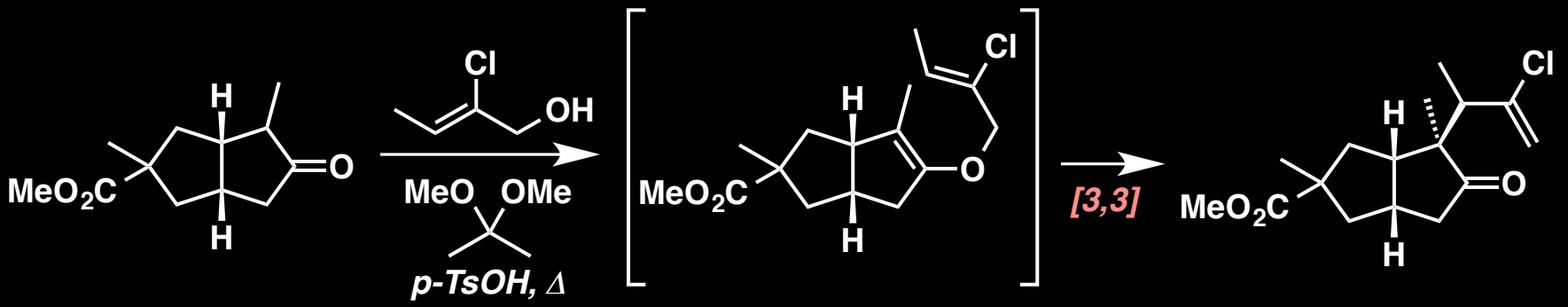
S. D. Burke and co-workers, Strategies and Tactics in Organic Synthesis 1987, 2, 57.

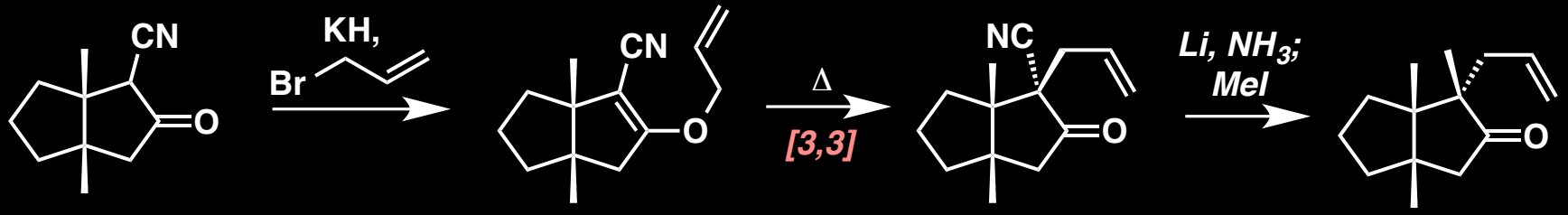
The Claisen Reaction: Substrate Preparation



P. T. Lansbury, N. Y. Wang, J. E. Rhodes, Tetrahedron Lett. 1971,12, 1829 and 1833
F. Johnson, K. G. Paul, D. Favaro, R. Ciabatti, U. Guzzi, J. Am. Chem. Soc. 1982, 104, 2198.

The Claisen Reaction: Substrate Preparation

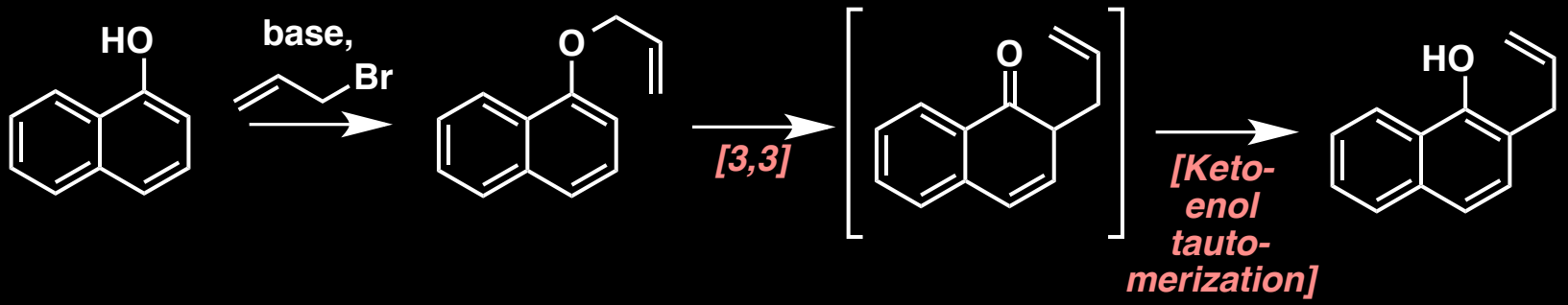




In both cases the Claisen reaction occurs on the more accessible, convex face of the molecule

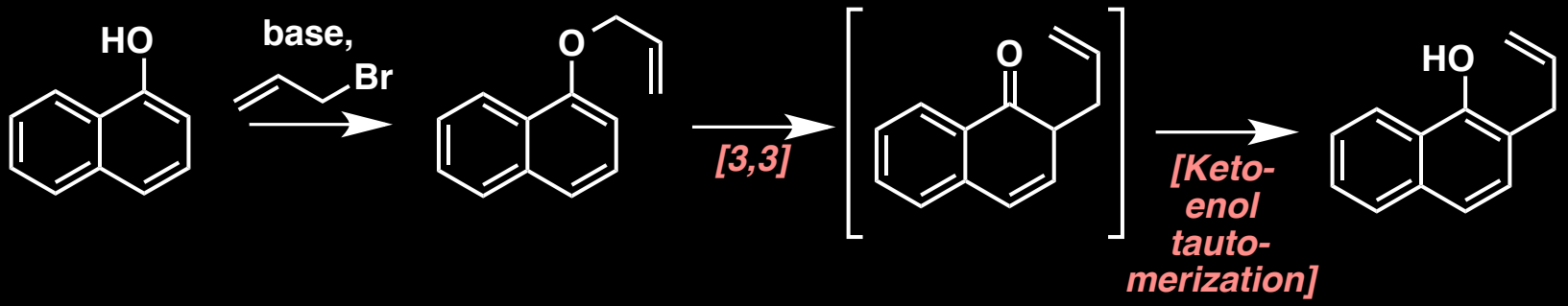
P. T. Lansbury, N. Y. Wang, J. E. Rhodes, Tetrahedron Lett. 1971,12, 1829 and 1833
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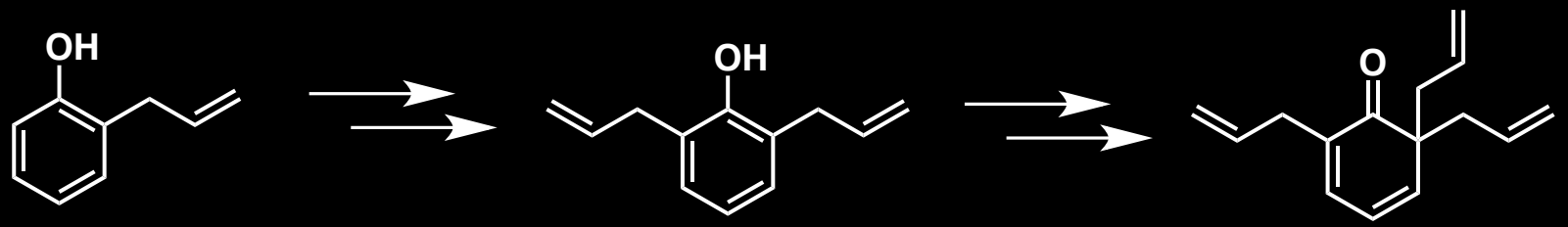
The Claisen Reaction: Substrate Preparation



Aromaticity can be broken through the Claisen rearrangement

The Claisen Reaction: Substrate Preparation

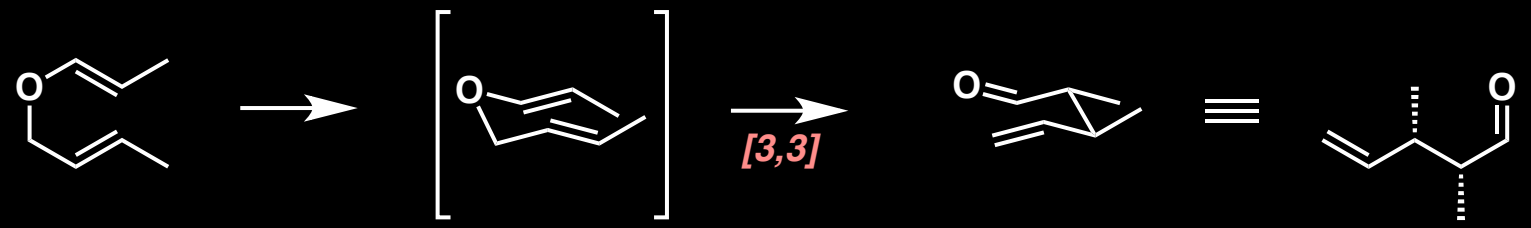




Aromaticity can be broken through the Claisen rearrangement

The Claisen Reaction: Stereochemical Considerations

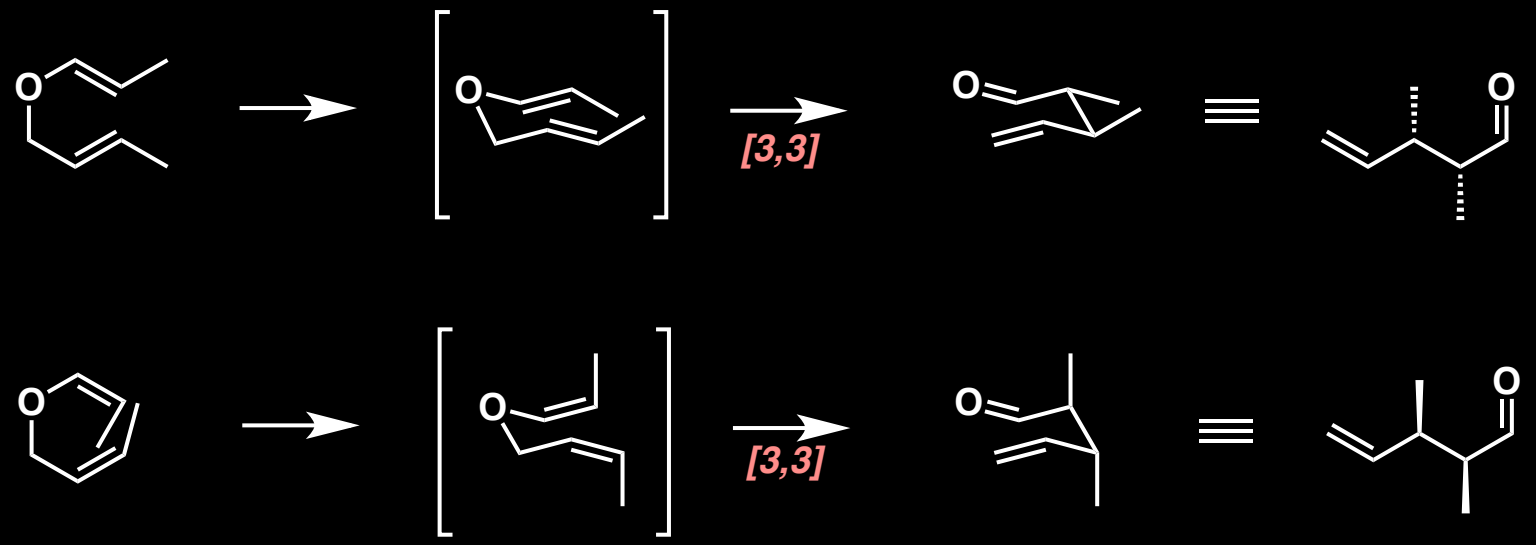
Reactions proceed through chair-like transition states



P. A. Bartlett, Tetrahedron 1980, 36, 2.

The Claisen Reaction: Stereochemical Considerations

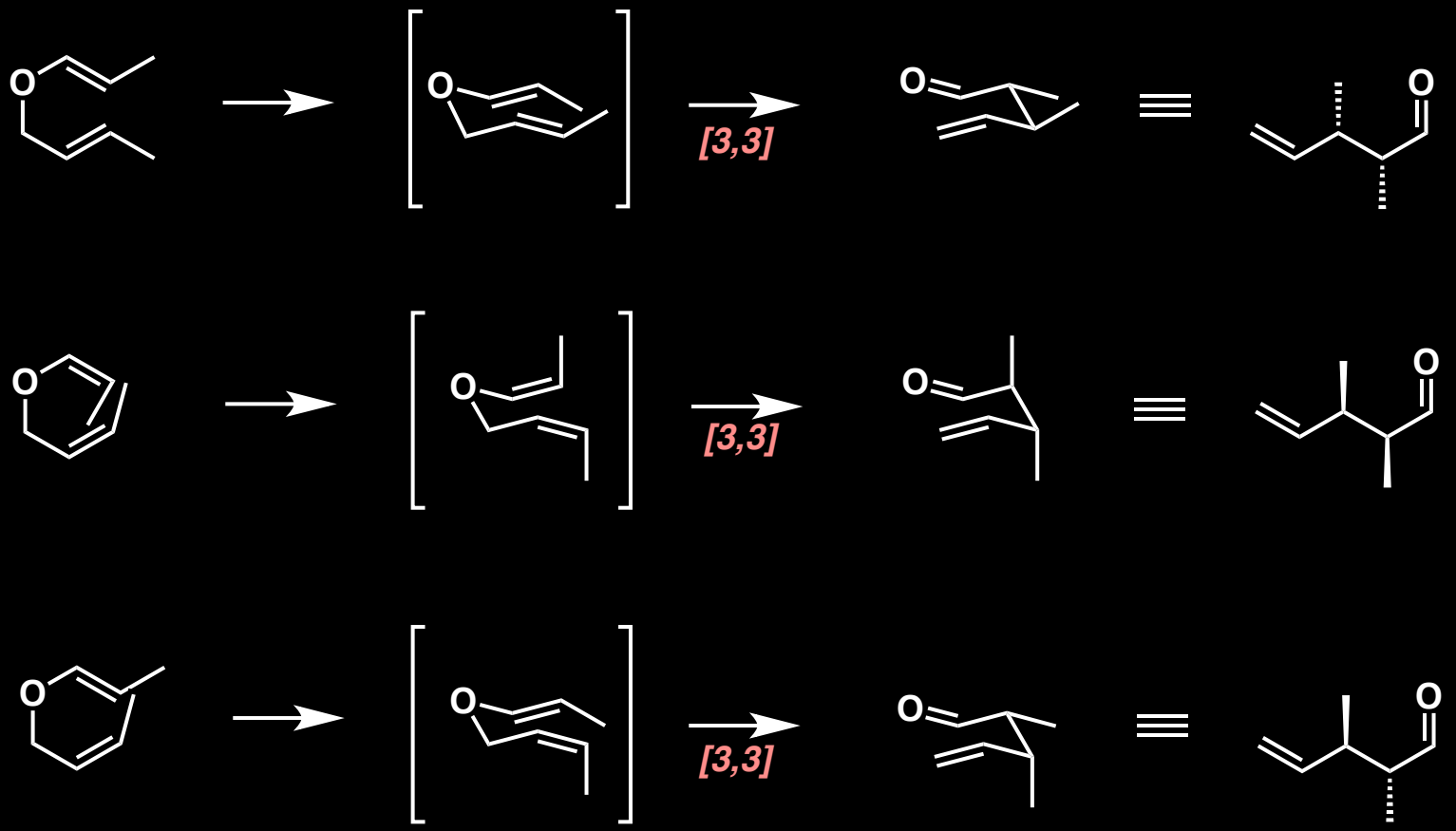
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The Claisen Reaction: Stereochemical Considerations

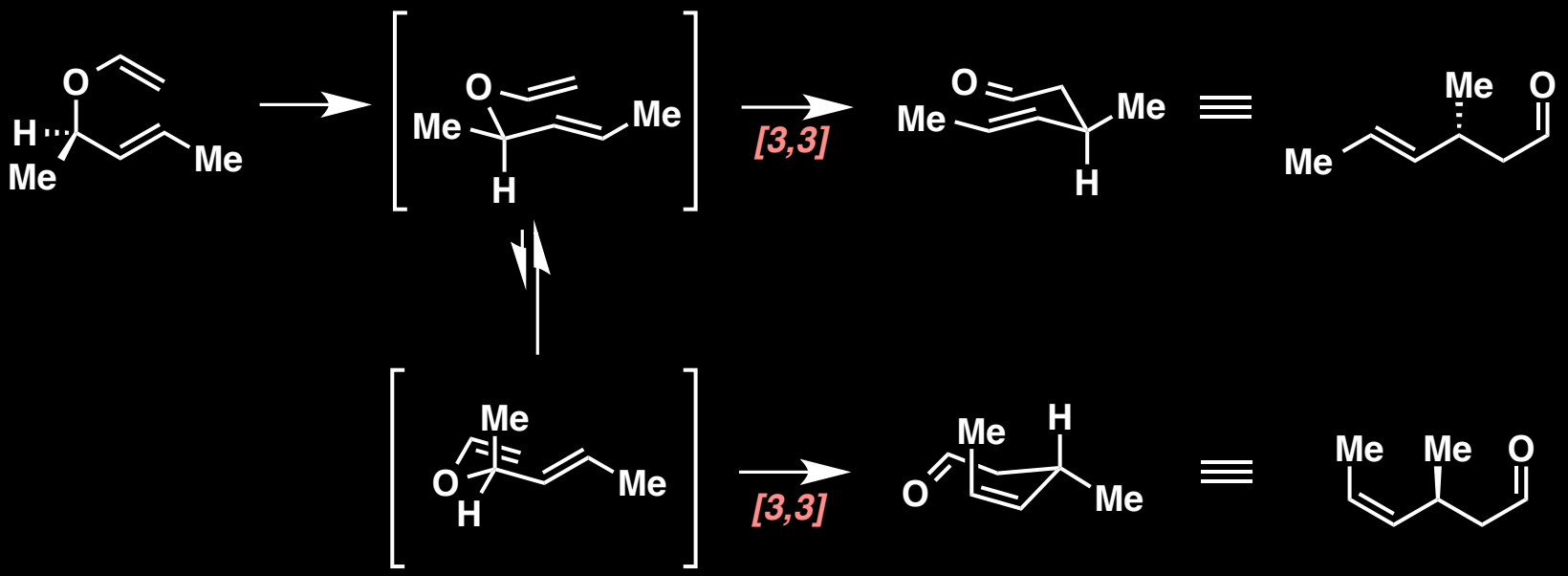
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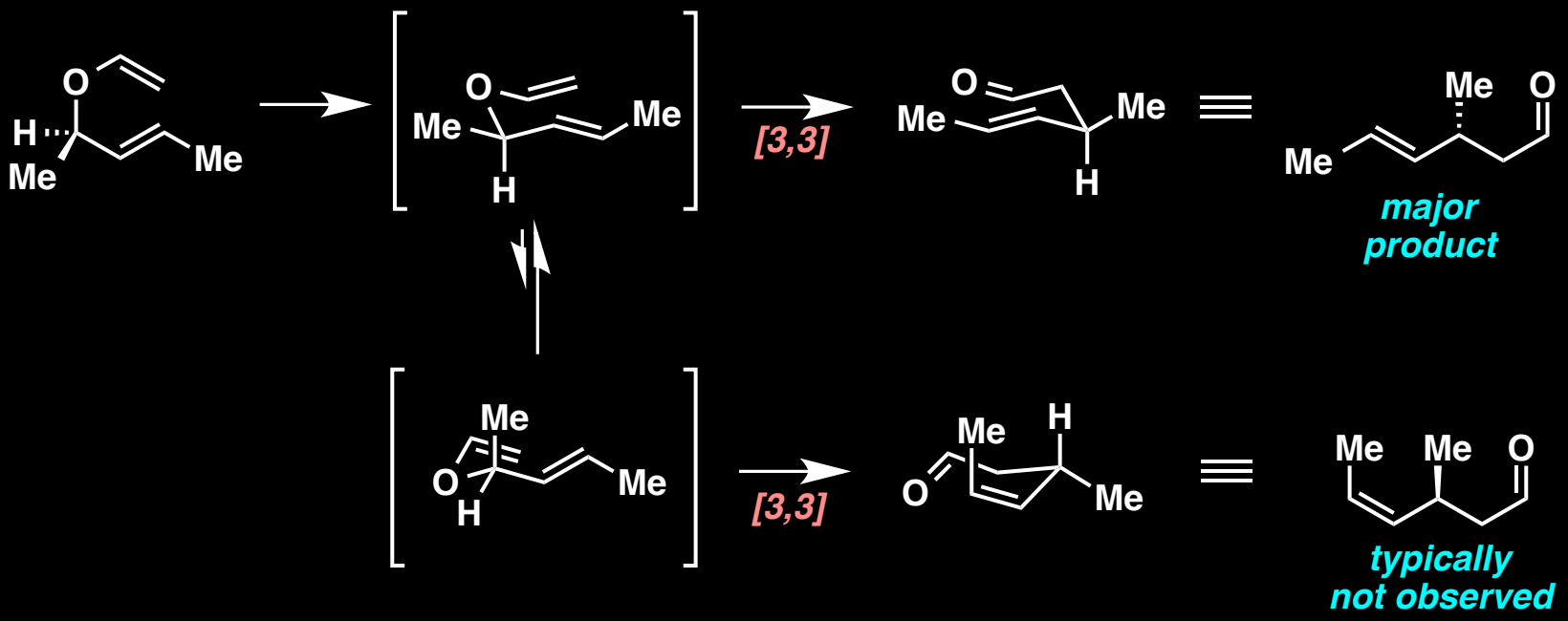
The Claisen Reaction: Stereochemical Considerations

Pre-existing chirality determines the stereoselectivity of the rearrangement



The Claisen Reaction: Stereochemical Considerations

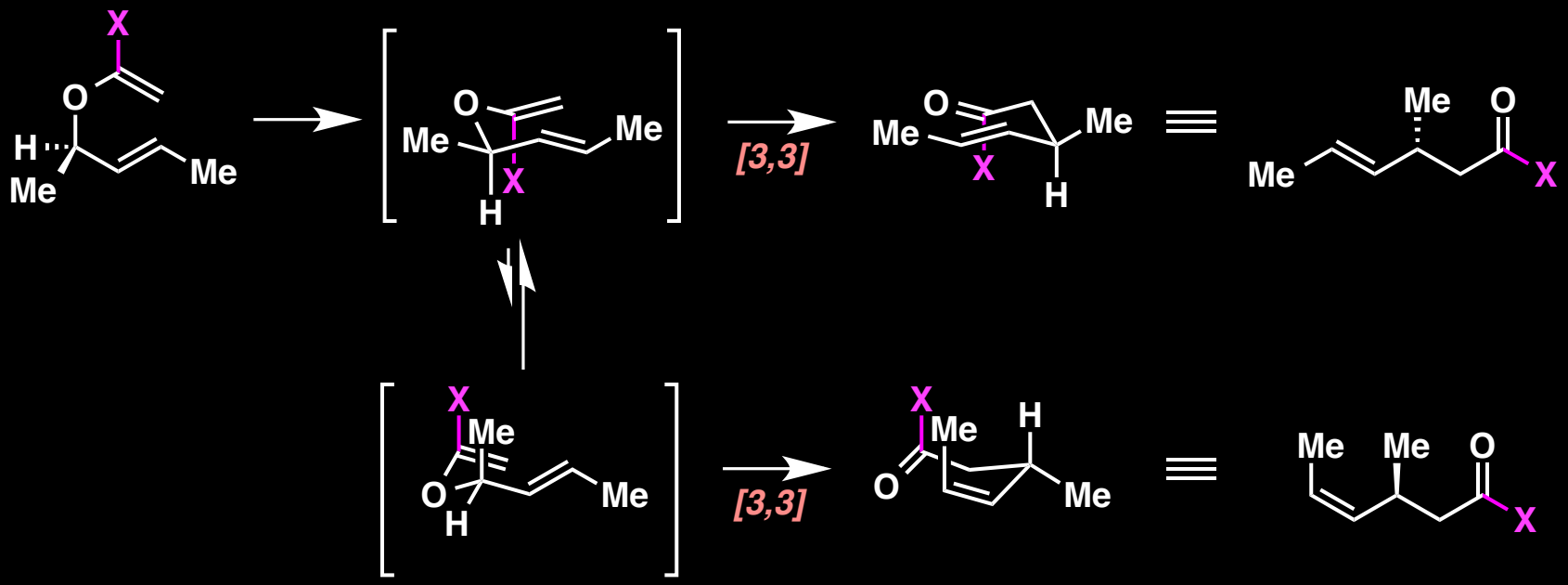
Pre-existing chirality determines the stereoselectivity of the rearrangement



As a general rule, unless there are strange geometrical constraints (i.e. the reaction proceeds in a chair, not a boat, transition state), E-olefins result from the Claisen rearrangement

The Claisen Reaction: Stereochemical Considerations

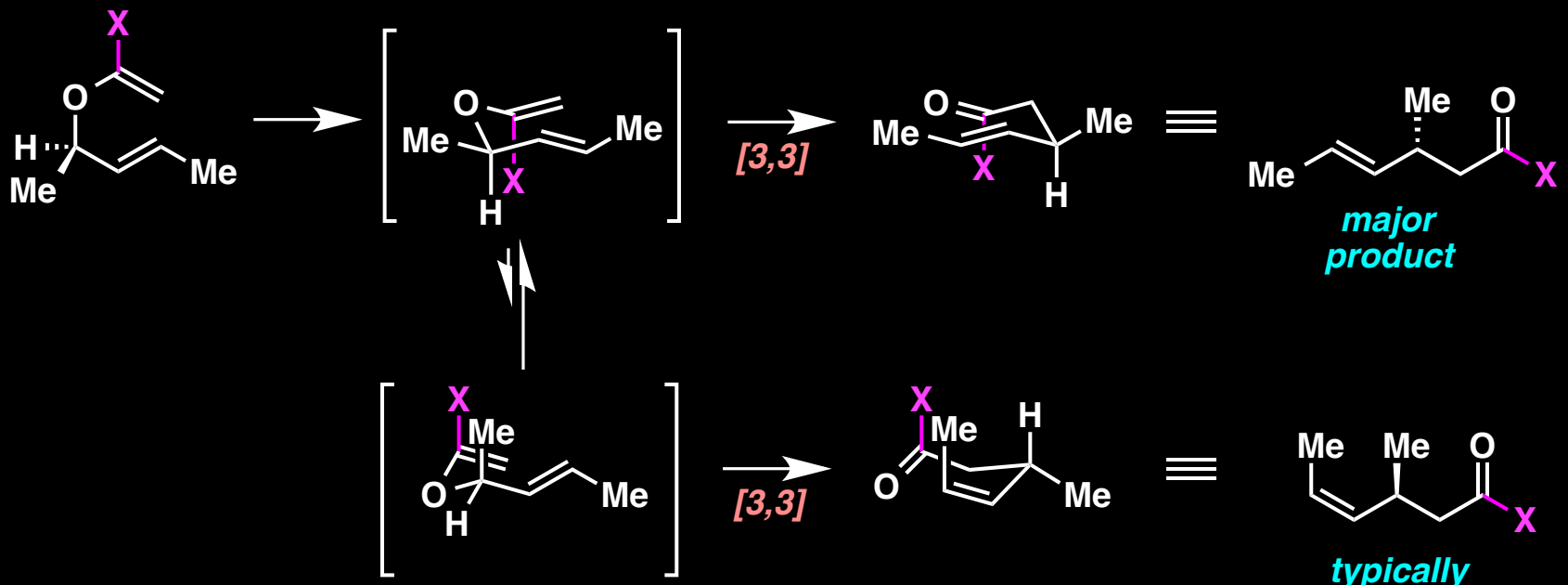
Pre-existing chirality determines the stereoselectivity of the rearrangement



C. L. Perrin, D. J. Faulkner, Tetrahedron Lett. 1969, 2783.

The Claisen Reaction: Stereochemical Considerations

Pre-existing chirality determines the stereoselectivity of the rearrangement



X = H, E/Z = ~9:1

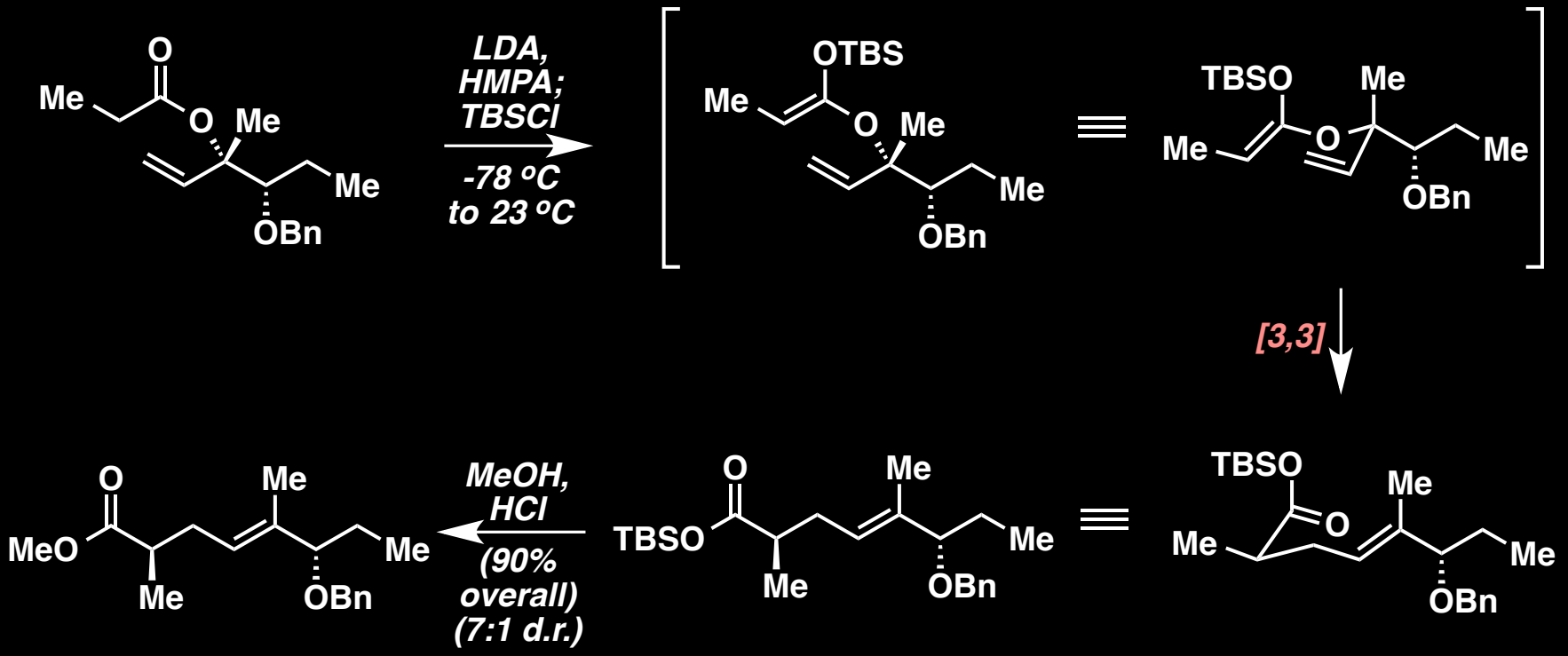
X = OEt or NMe_2, E/Z = >99:1

Preference for equatorial position of substituents correlates with A-values

C. L. Perrin, D. J. Faulkner, Tetrahedron Lett. 1969, 2783.

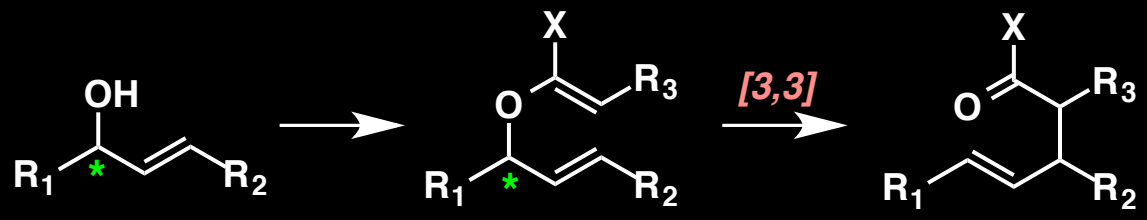
The Claisen Reaction: Stereochemical Considerations

Pre-existing chirality determines the stereoselectivity of the rearrangement, even if multiple substituents are present

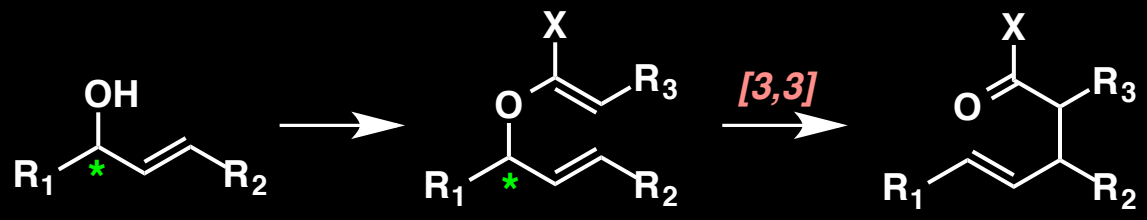


A. S. Franklin, L. E. Overman, Chem. Rev. 1996, 96, 505

The Claisen Reaction: Stereochemical Considerations



The Claisen Reaction: Stereochemical Considerations



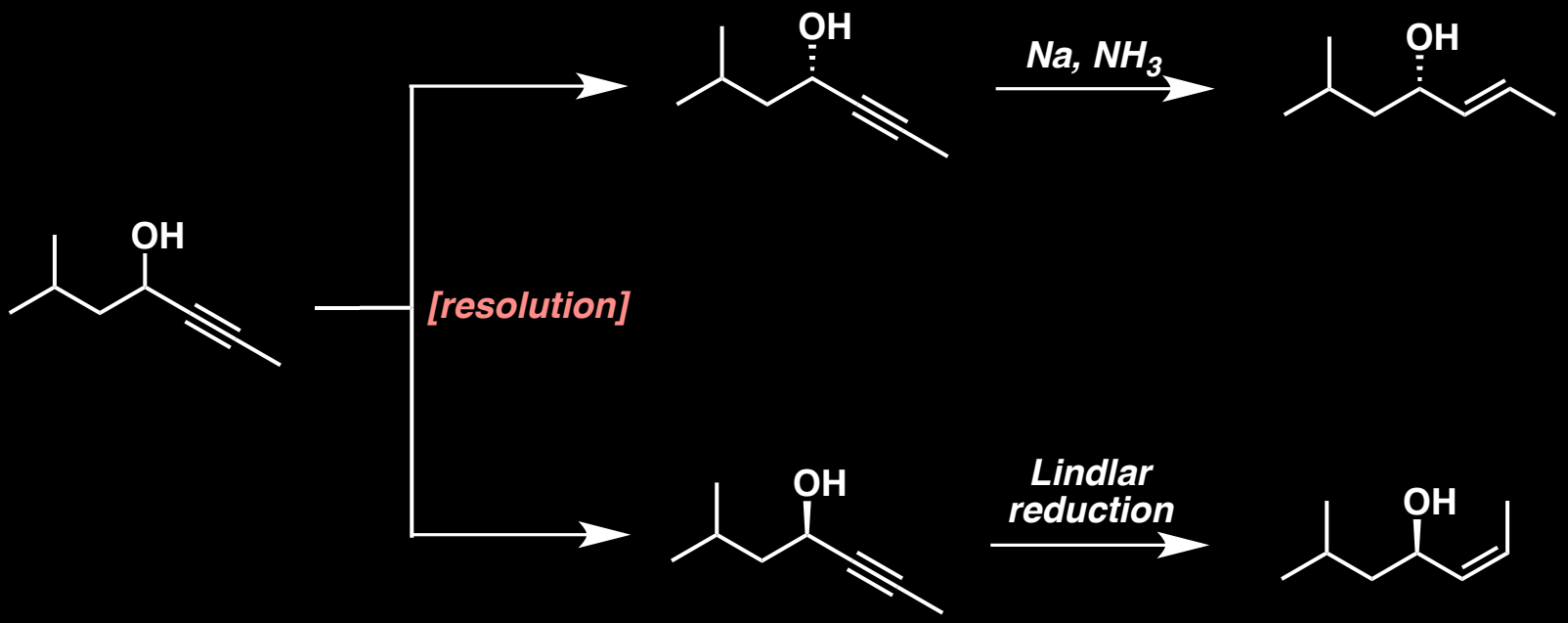
Many different ways to access chiral allylic alcohols:

Chiral pool building block

Resolution of achiral starting material

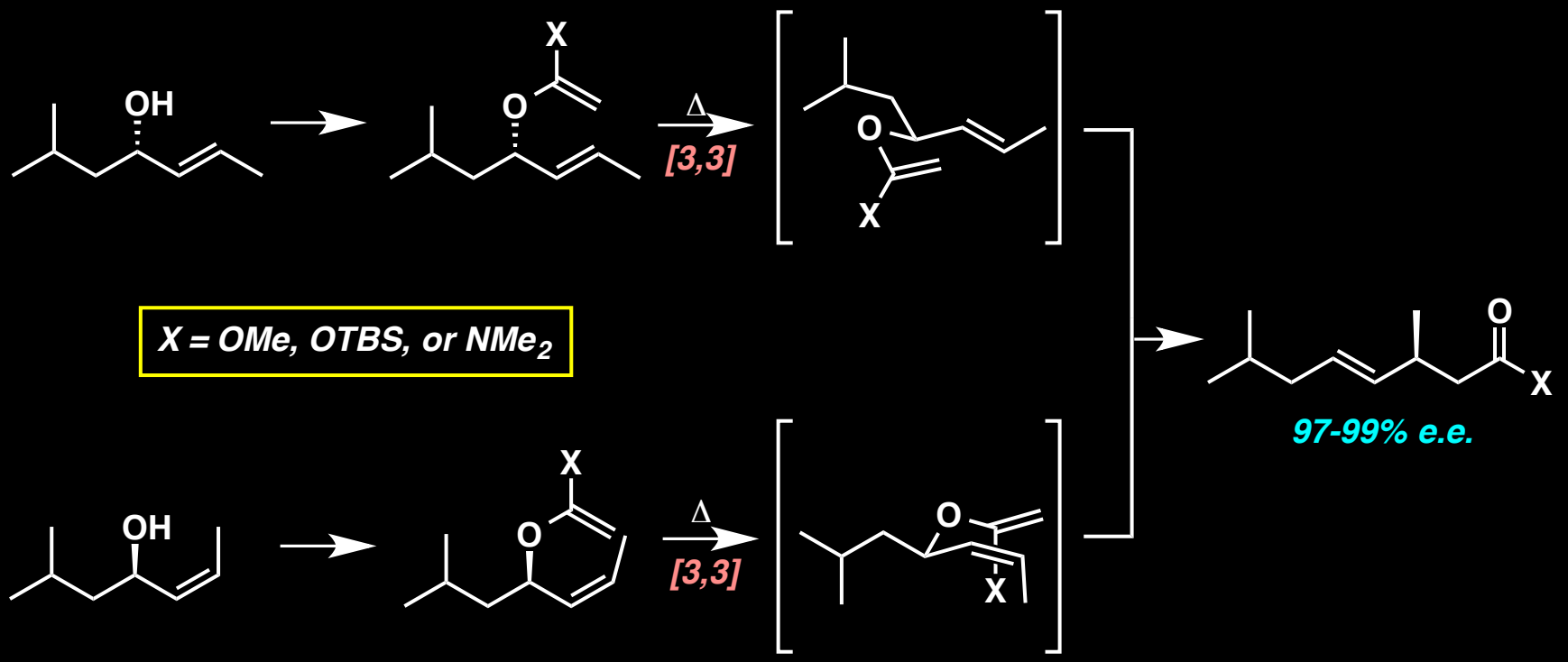
Asymmetric reduction of α,β-unsaturated system

The Roche Synthesis of α-Tocopherol



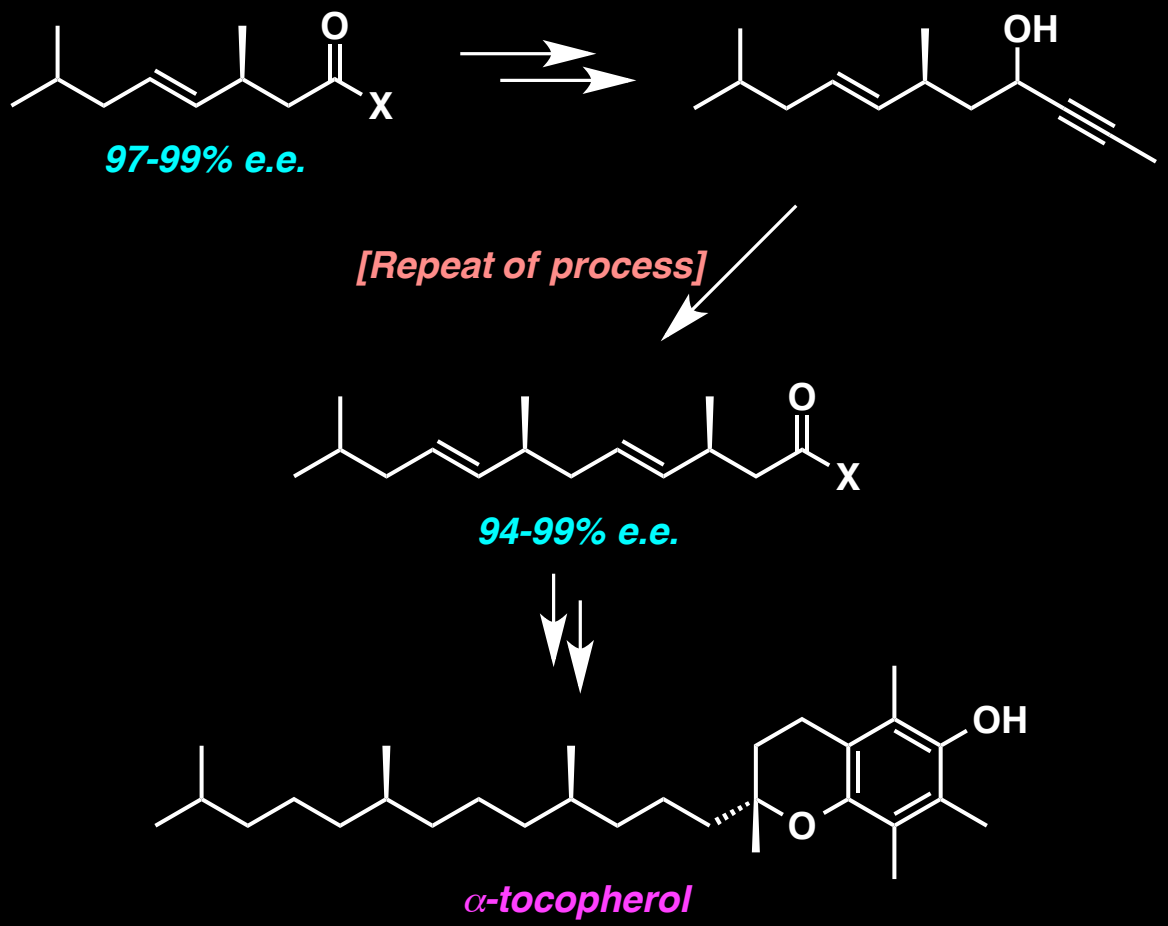
J. Org. Chem. 1976, 41, 3497; J. Org. Chem. 1976, 41, 3512; J. Org. Chem. 1978, 43, 3435.

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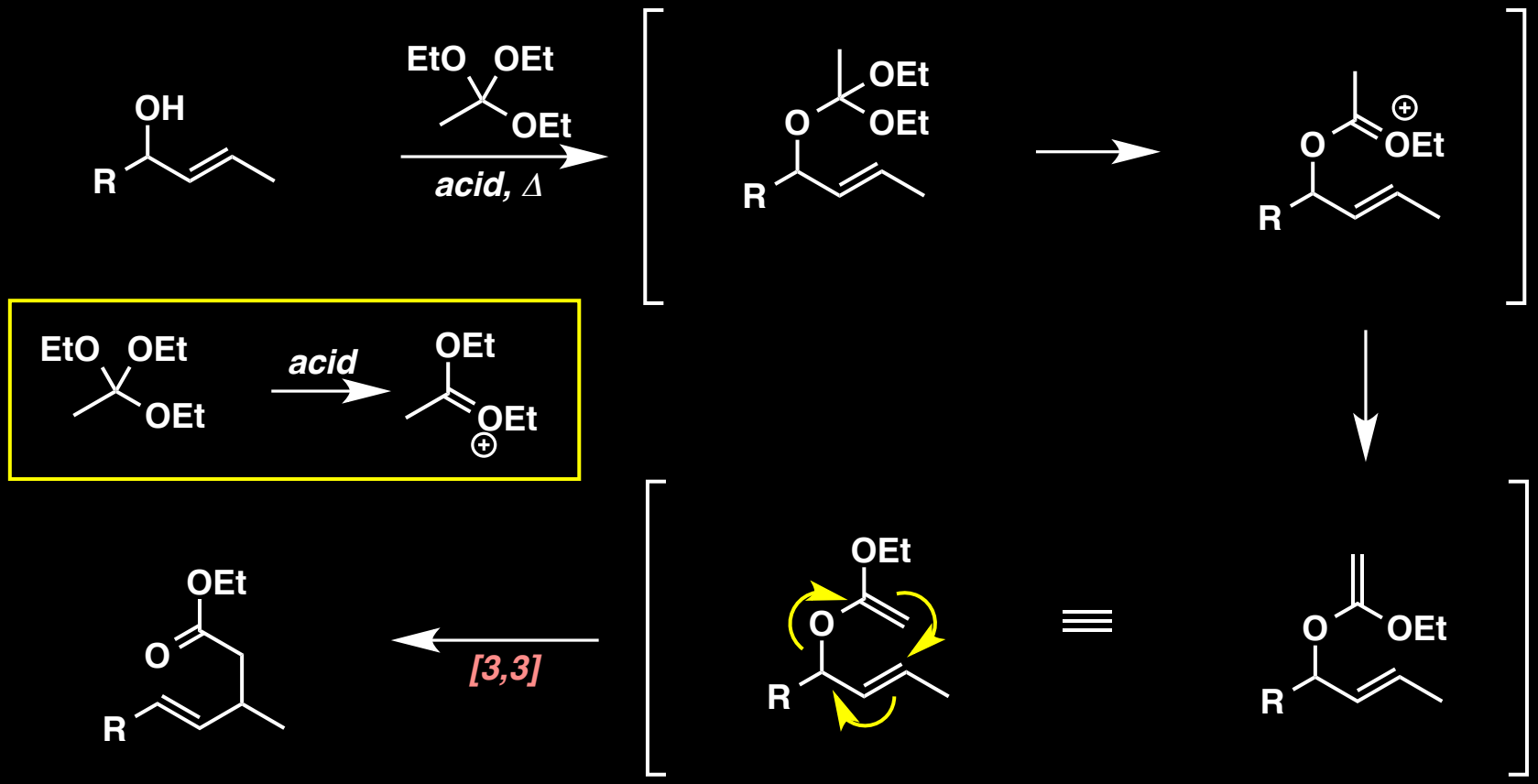
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J. Org. Chem. 1976, 41, 3497; J. Org. Chem. 1976, 41, 3512; J. Org. Chem. 1978, 43, 3435.

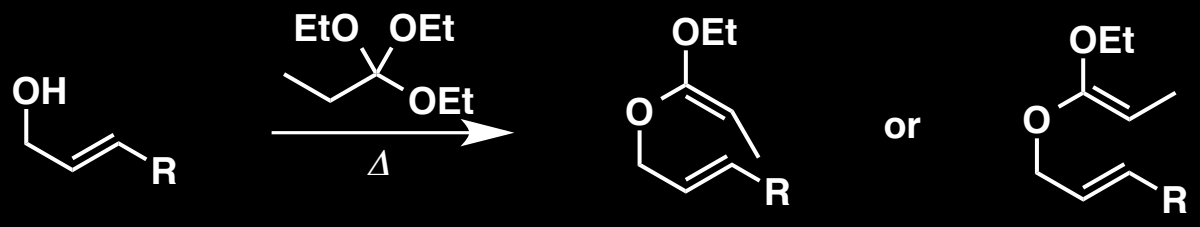
The Johnson Orthoester Claisen Rearrangement: A Useful Variant of the Claisen Reaction



W.S. Johnson, L. Werthemann, W.R. Bartless, T.J. Brocksom, T.-T. Li, D. J. Faulkner, M.R. Petersen, J. Am. Chem. Soc. 1970, 92, 741.

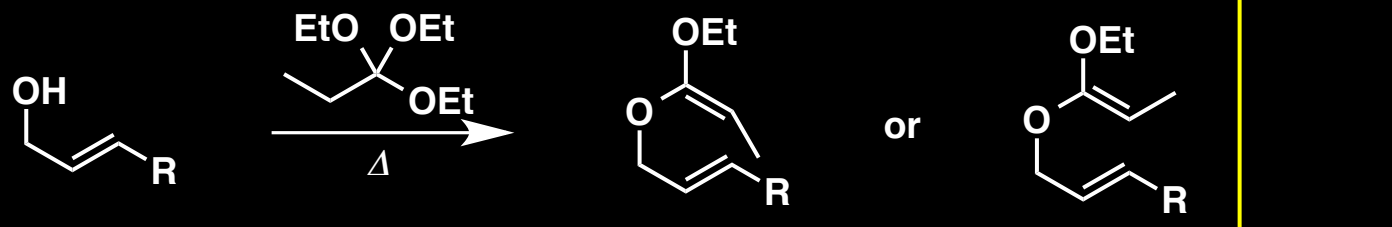
The Johnson Orthoester Claisen Rearrangement: Key Stereochemical Difference from Eschenmoser Variant

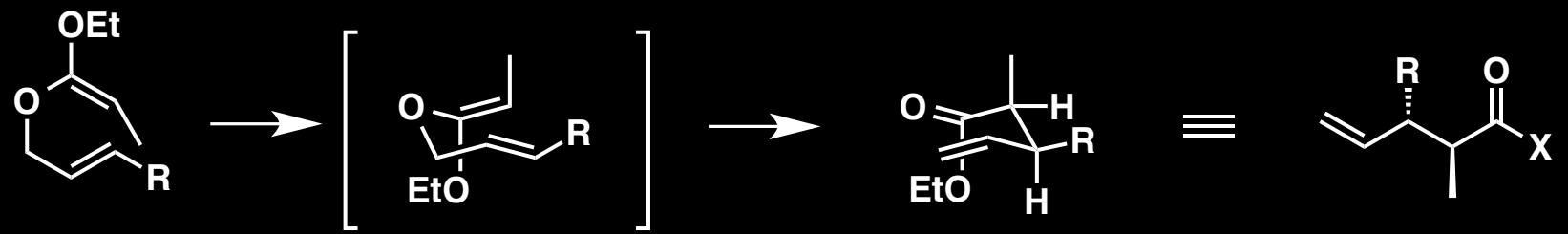
Stereocontrol can become a concern with non-methyl alkyl chains

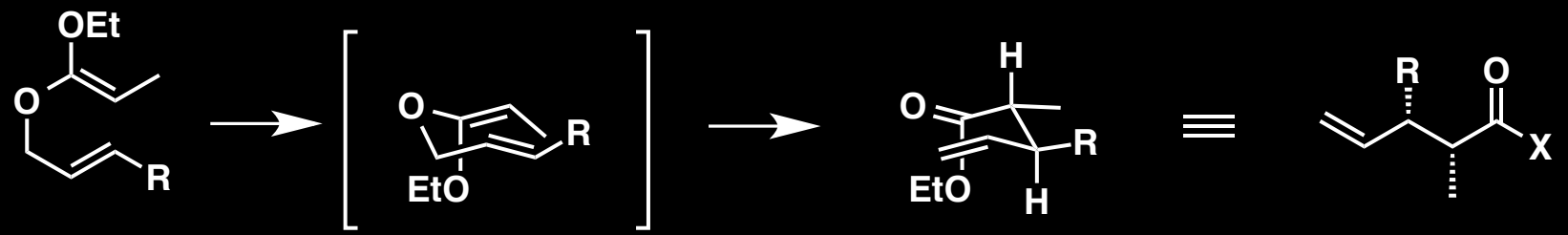


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Stereocontrol can become a concern with non-methyl alkyl chains

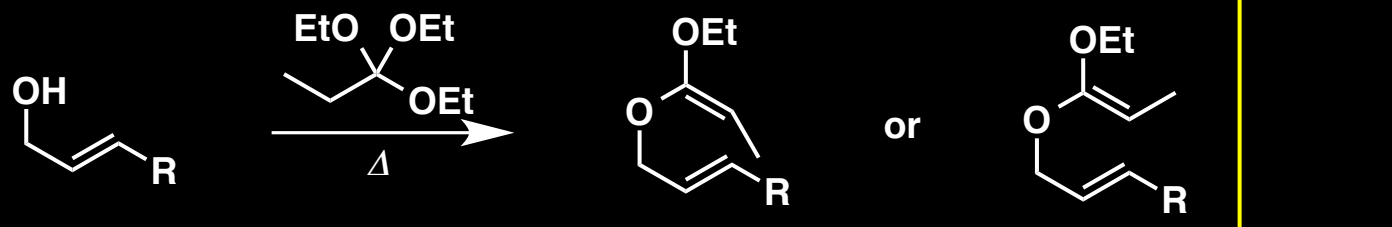


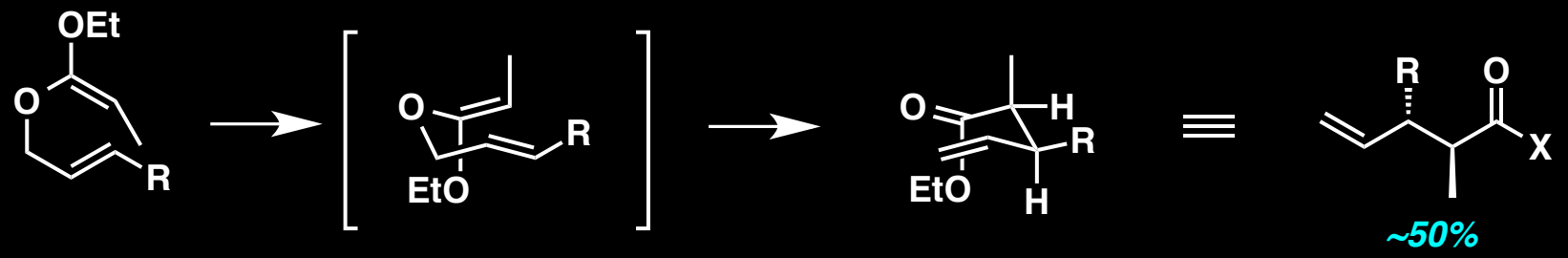


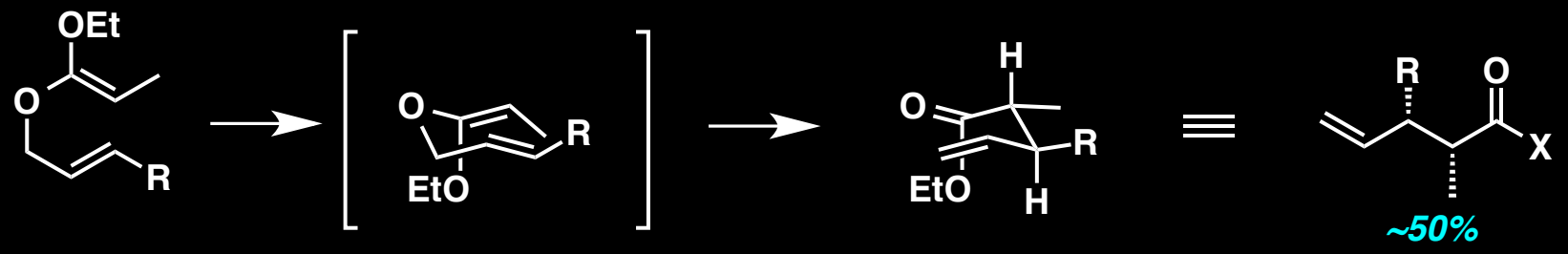


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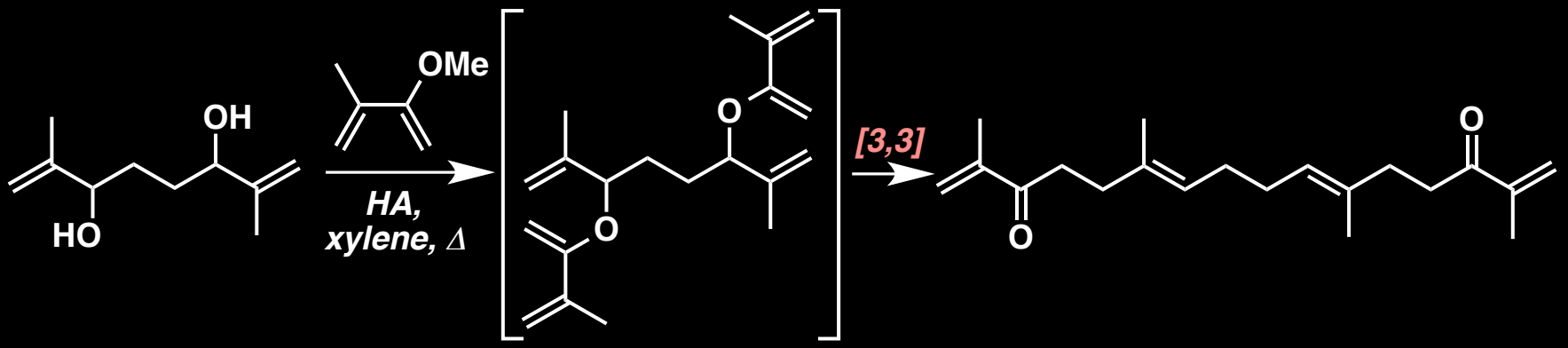
Stereocontrol can become a concern with non-methyl alkyl chains





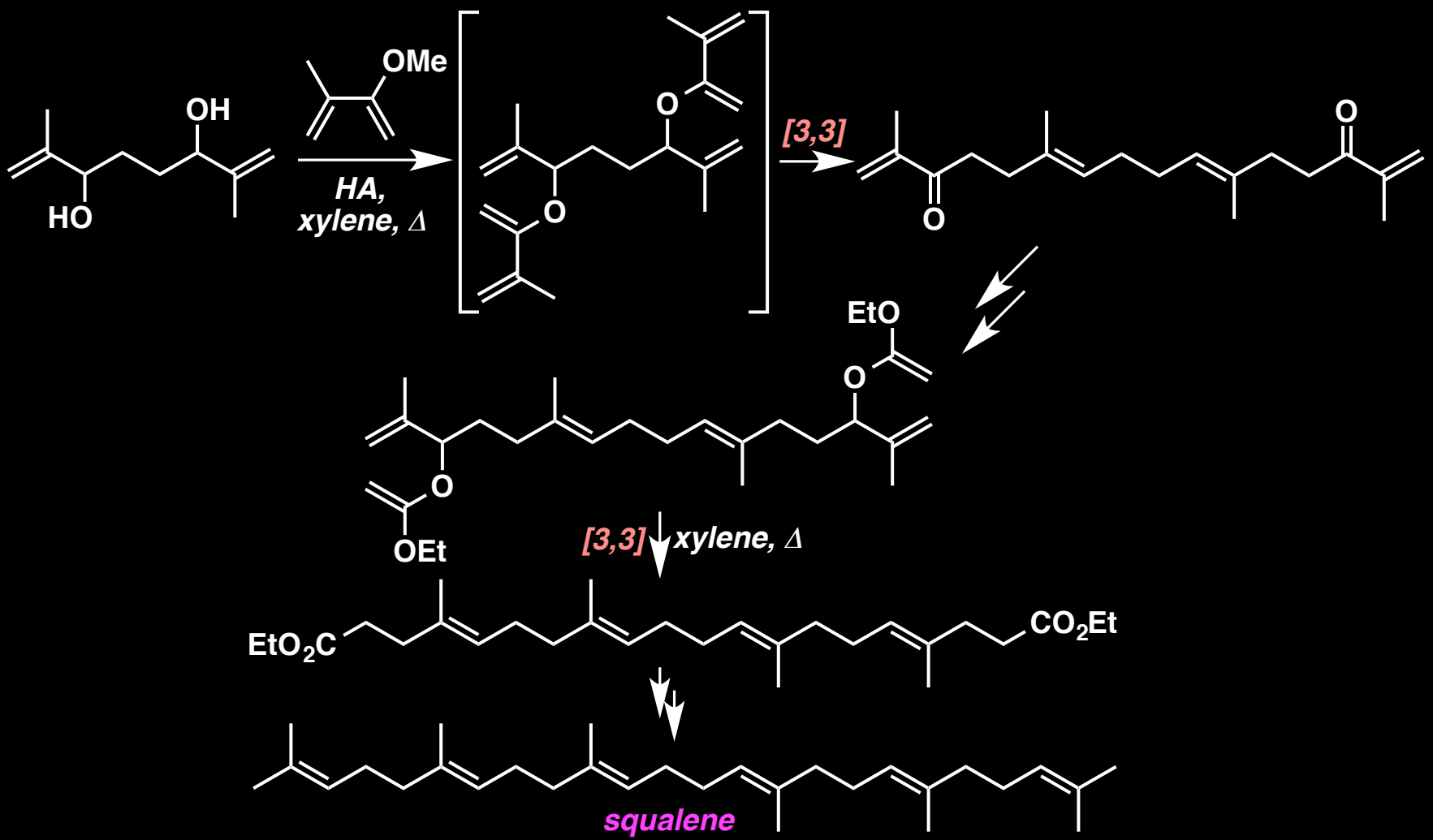


Johnson Orthoester Claisen Rearrangements in Synthesis



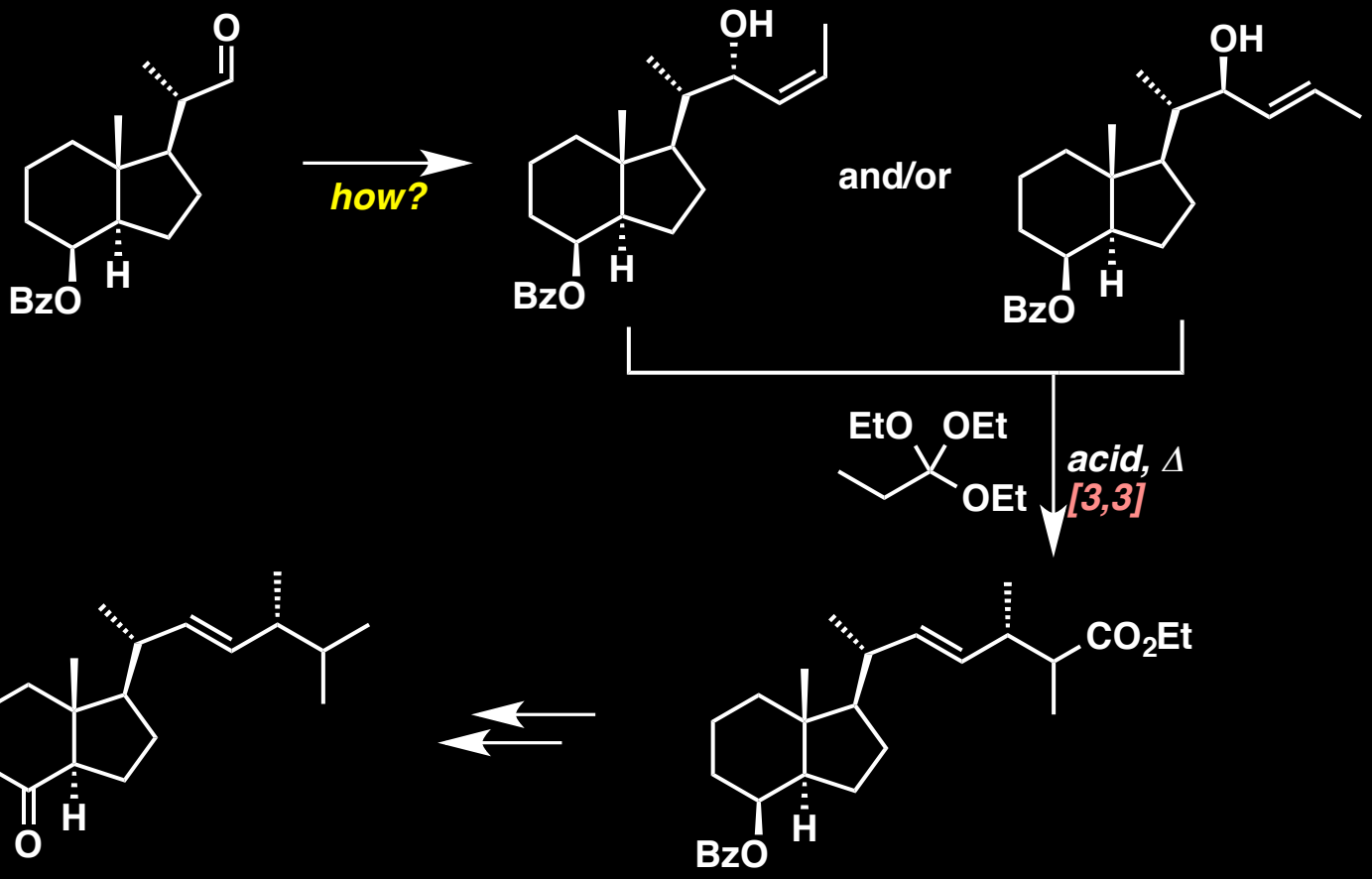
B. Lythgoe, D. A. Roberts, I. Waterhouse, J. Chem. Soc., Perkin Trans. 1 1977, 2608.

Johnson Orthoester Claisen Rearrangements in Synthesis



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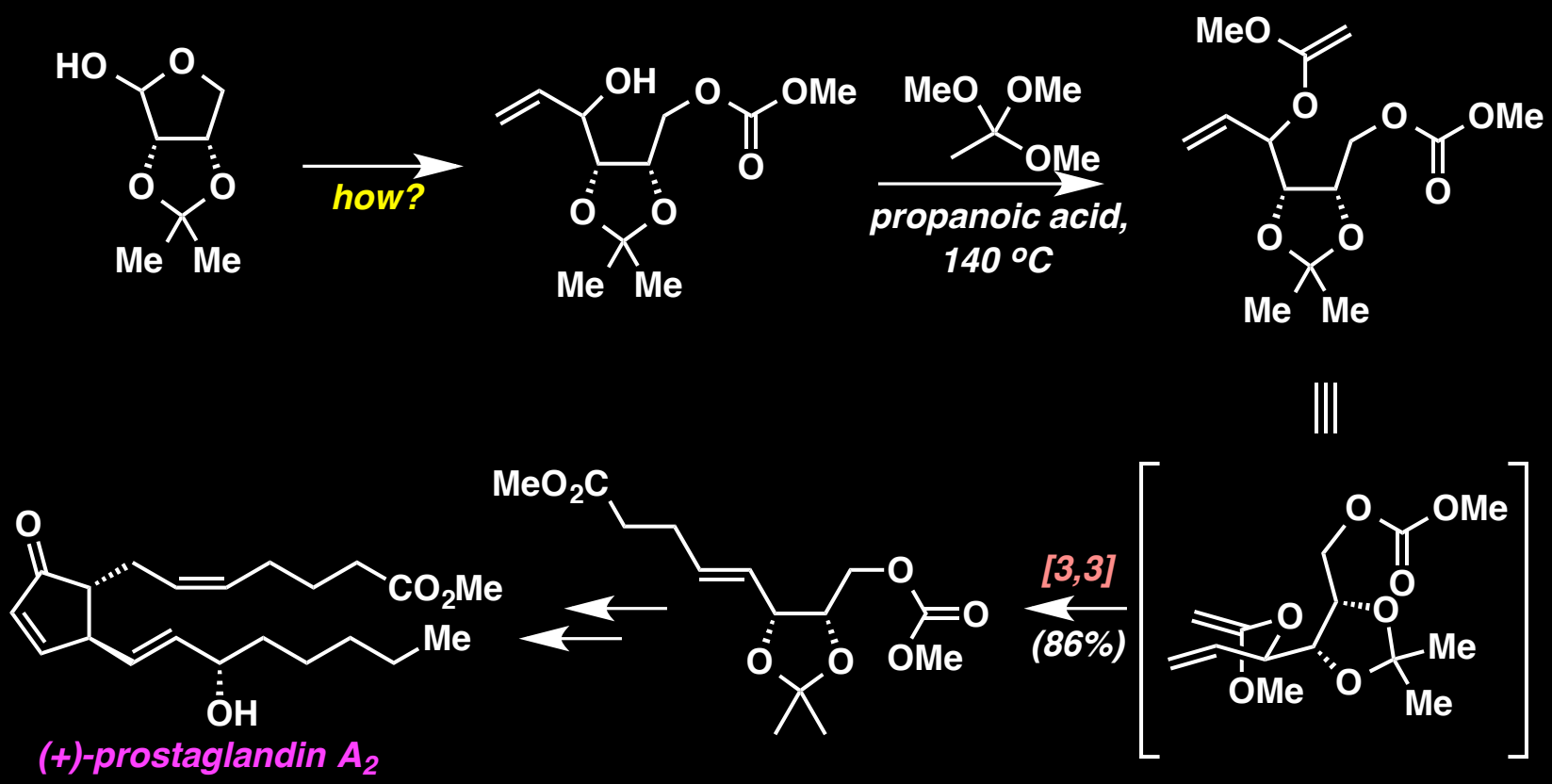
Johnson Orthoester Claisen Rearrangements in Synthesis



Windaus-Grundman ketone

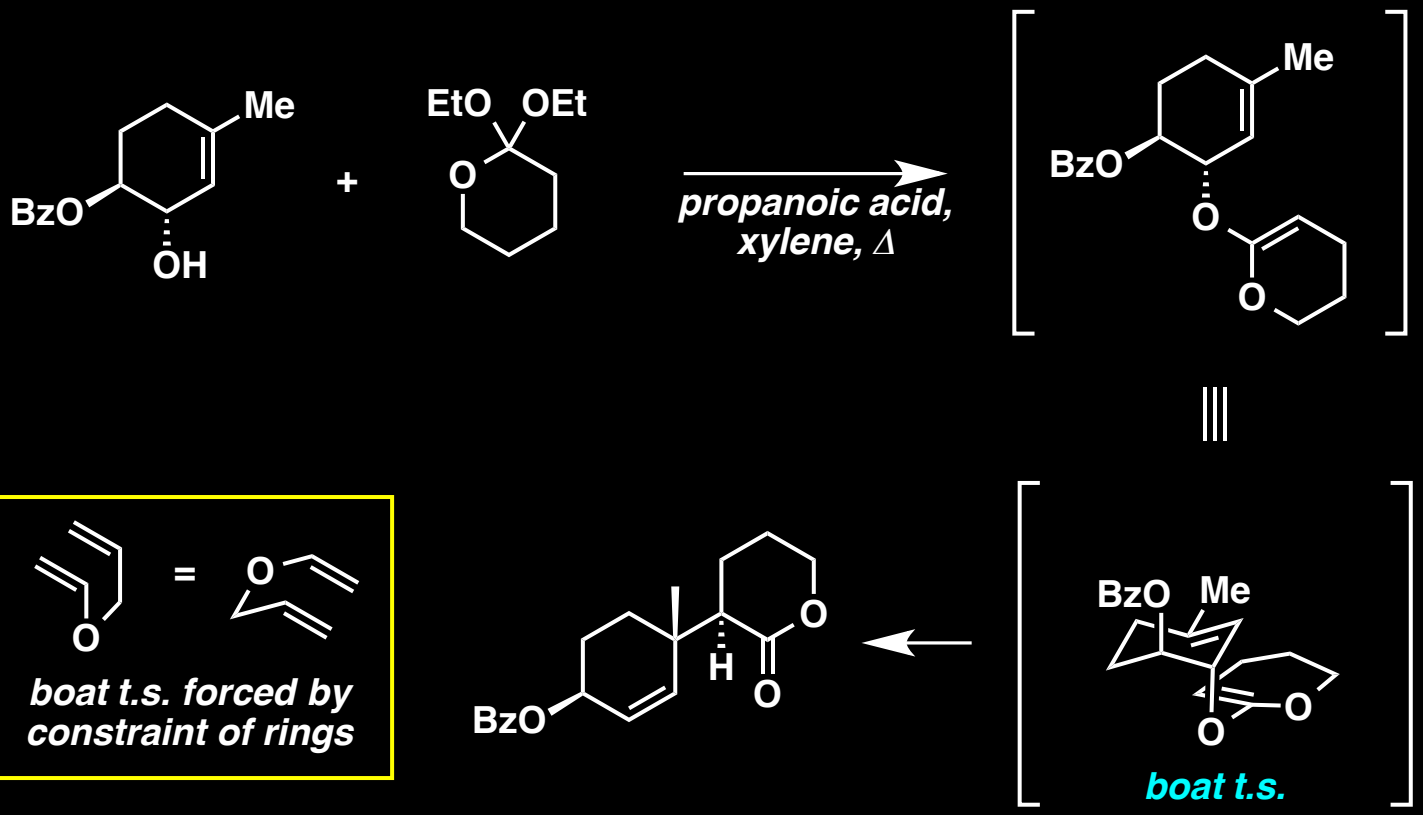
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Johnson Orthoester Claisen Rearrangements in Synthesis



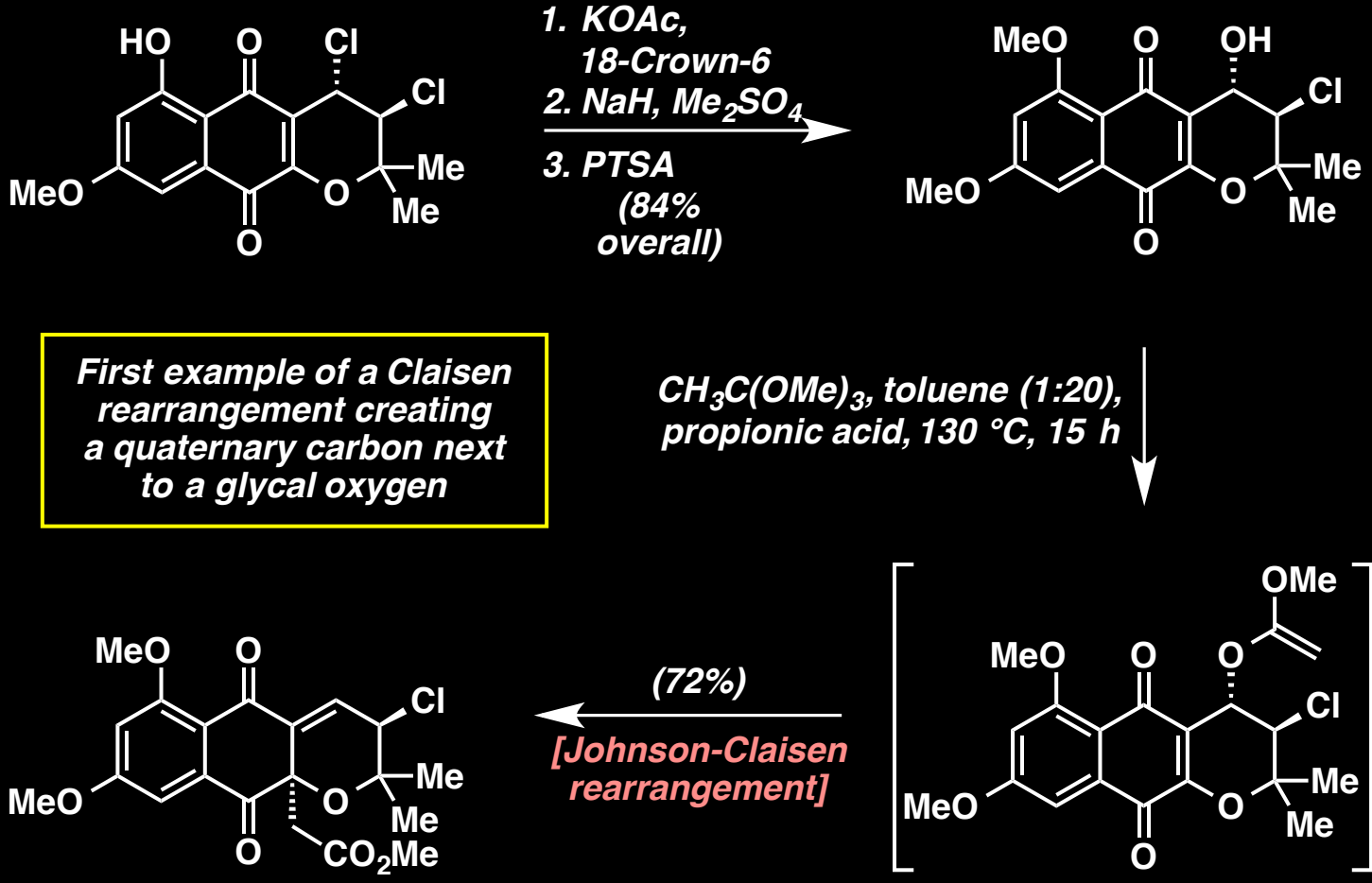
G. Stork, S. Raucher, J. Am. Chem. Soc. 1975, 98, 1583.

Johnson Orthoester Claisen Rearrangements in Synthesis



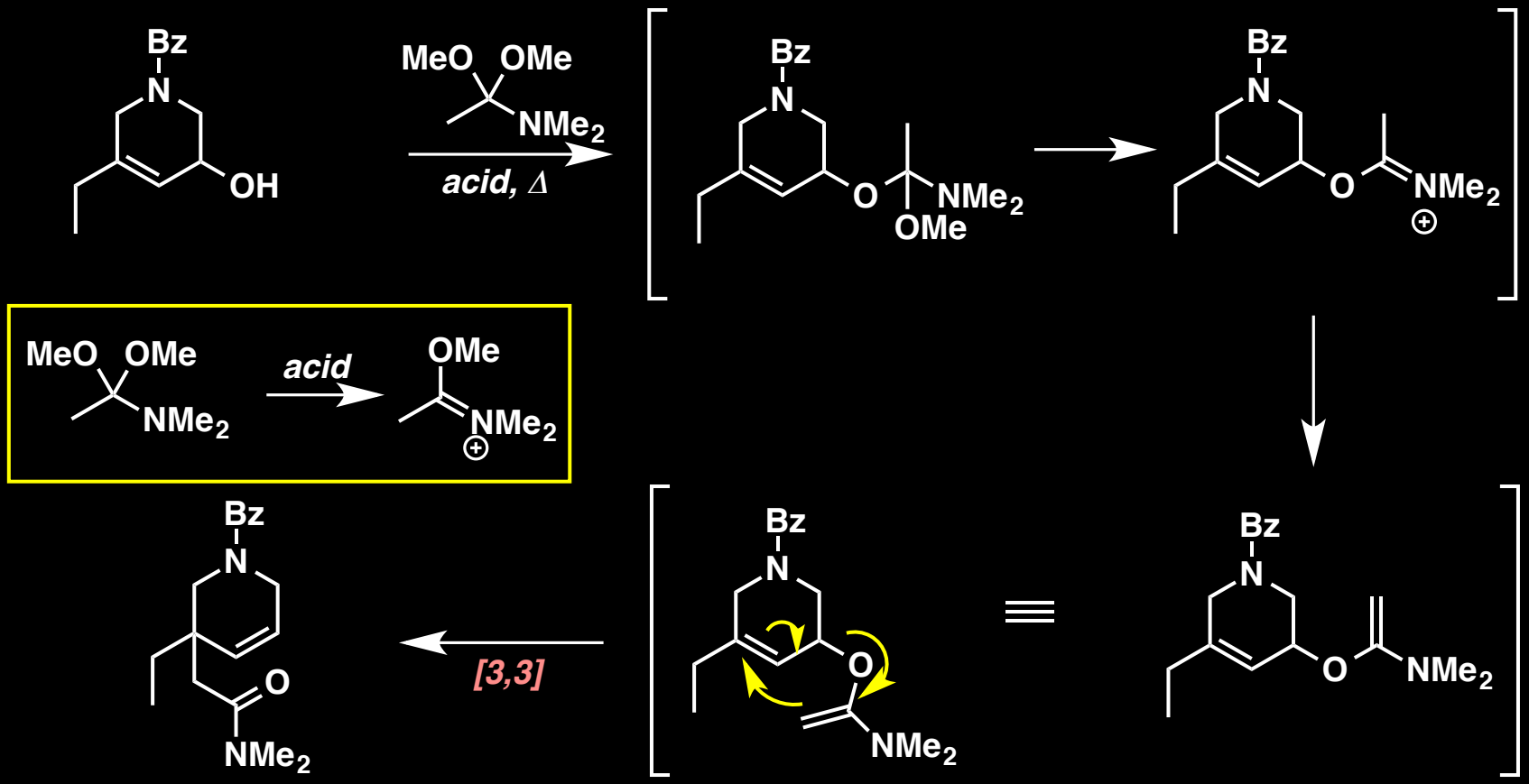
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Johnson Orthoester Claisen Rearrangements in Synthesis



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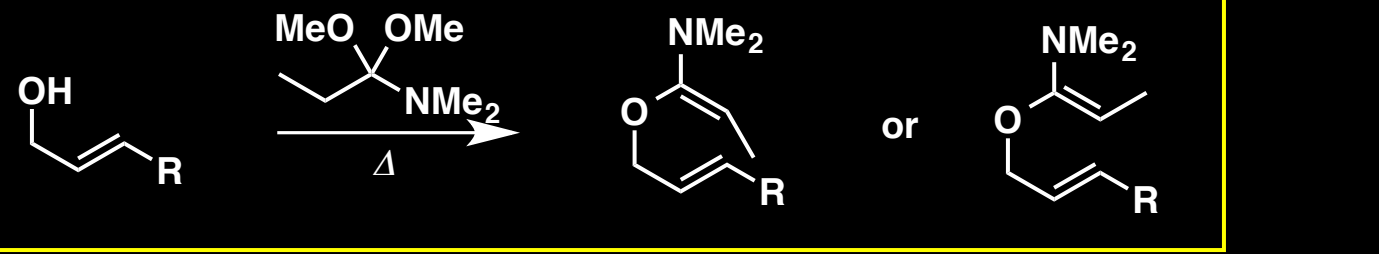
The Eschenmosher Amide Acetal Rearrangement: A Useful Variant of the Claisen Reaction

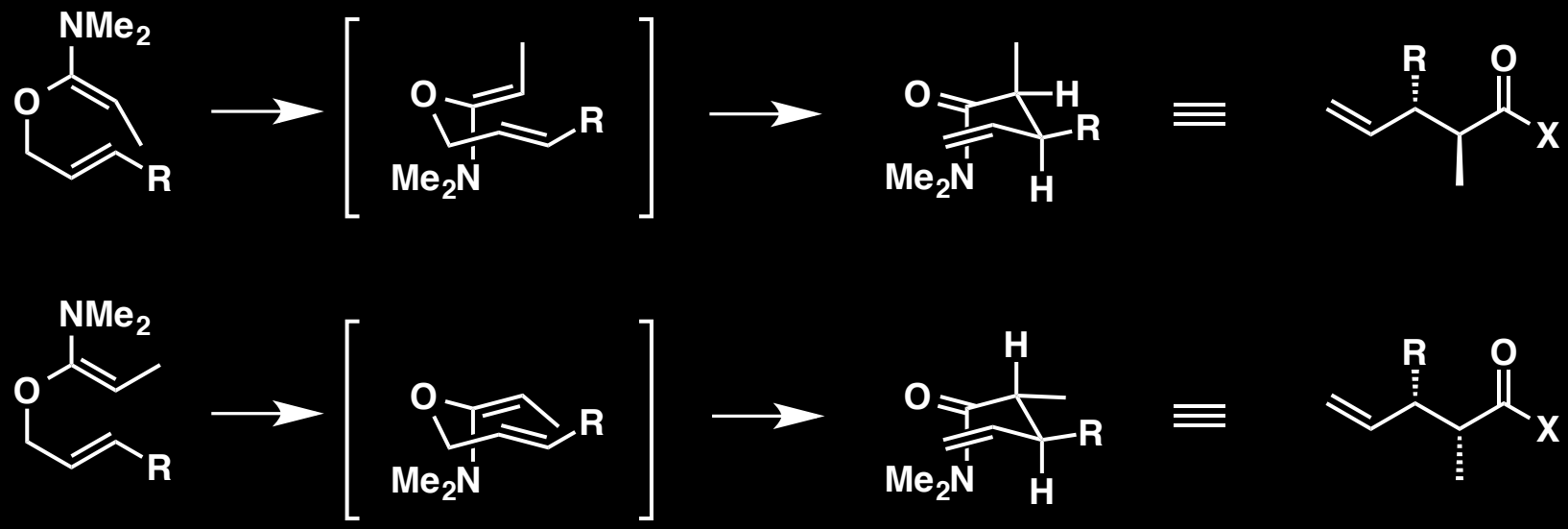


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The Eschenmosher Amide Acetal Rearrangement: A Useful Variant of the Claisen Reaction

Stereocontrol can become a concern with non-methyl alkyl chains

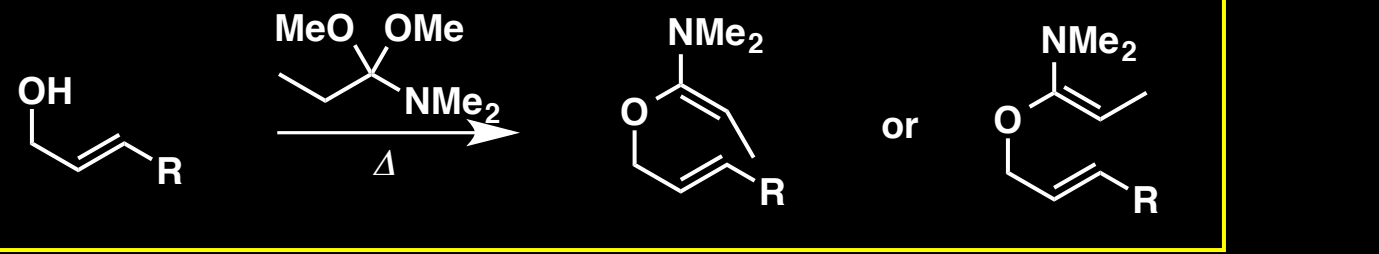


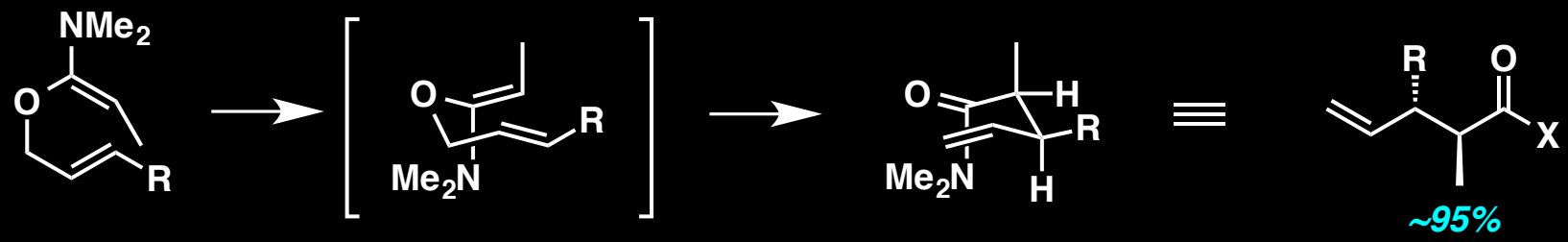


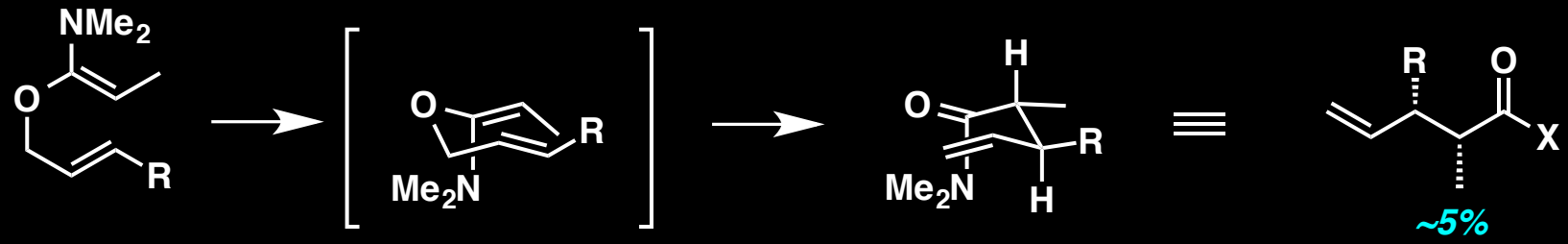
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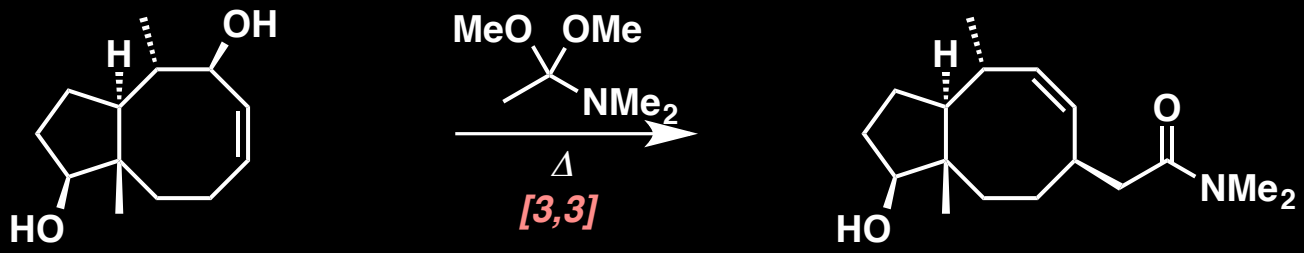


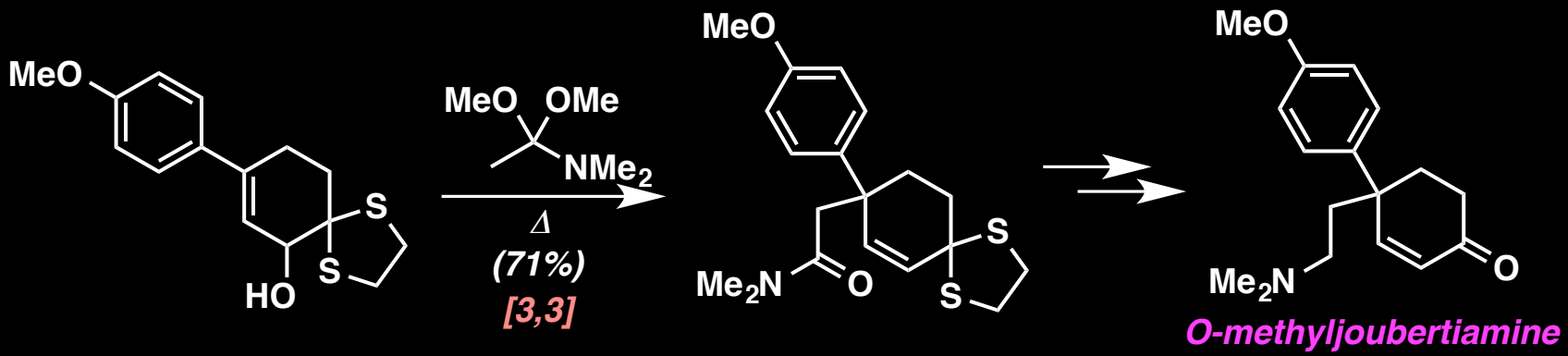




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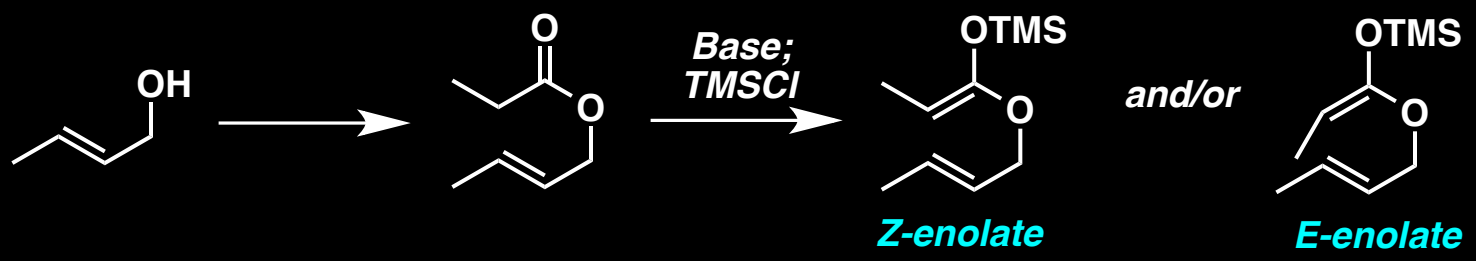




O-methyljoubertiamine

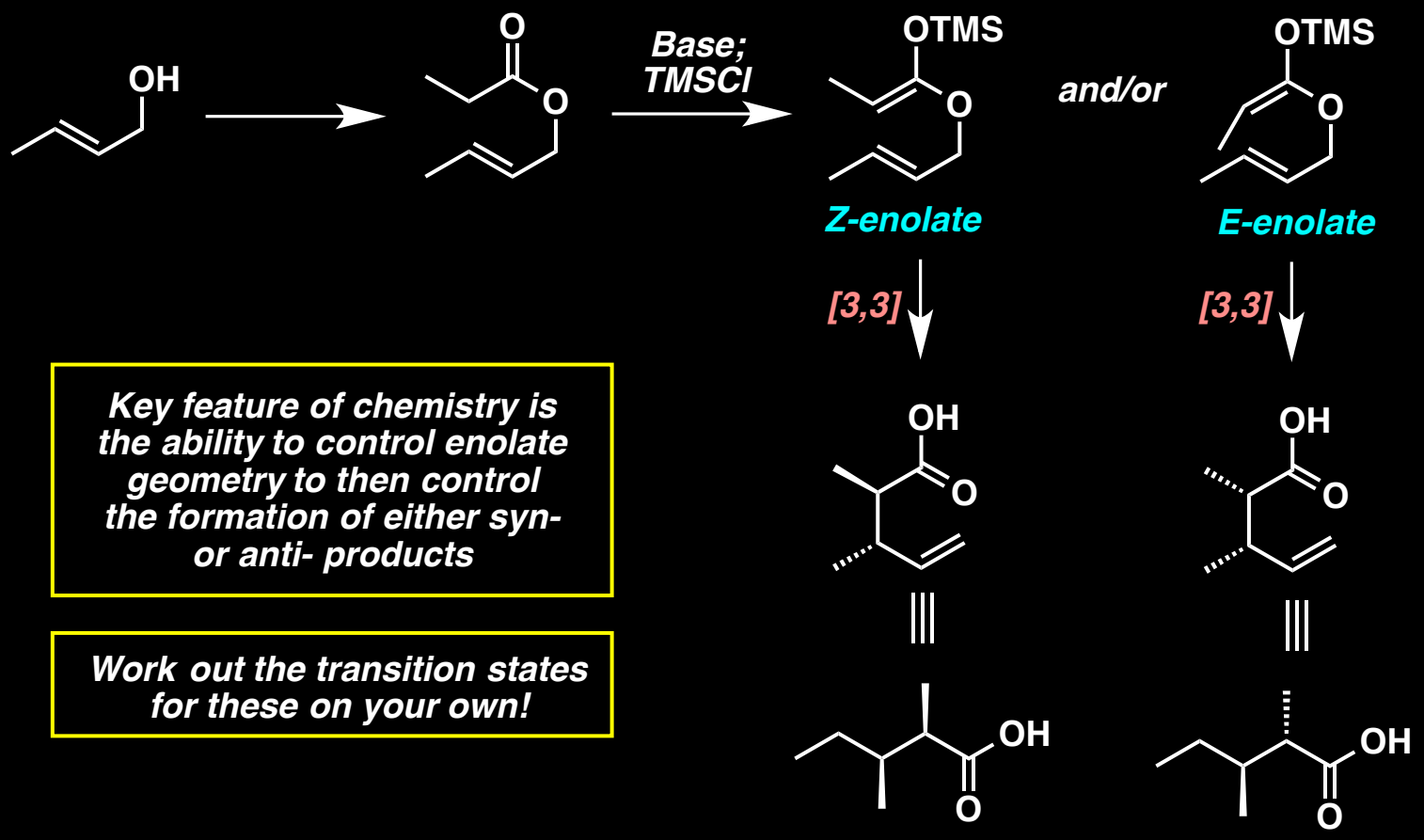
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The Ireland-Claisen Rearrangement: Use of Enolates to Generate Well-Defined Precursors



R. E. Ireland, R. H. Mueller, J. Am. Chem. Soc. 1972, 94, 5897.
R. E. Ireland, R. H. Mueller, A. K. Willard, J. Am. Chem. Soc. 1976, 98, 2868.

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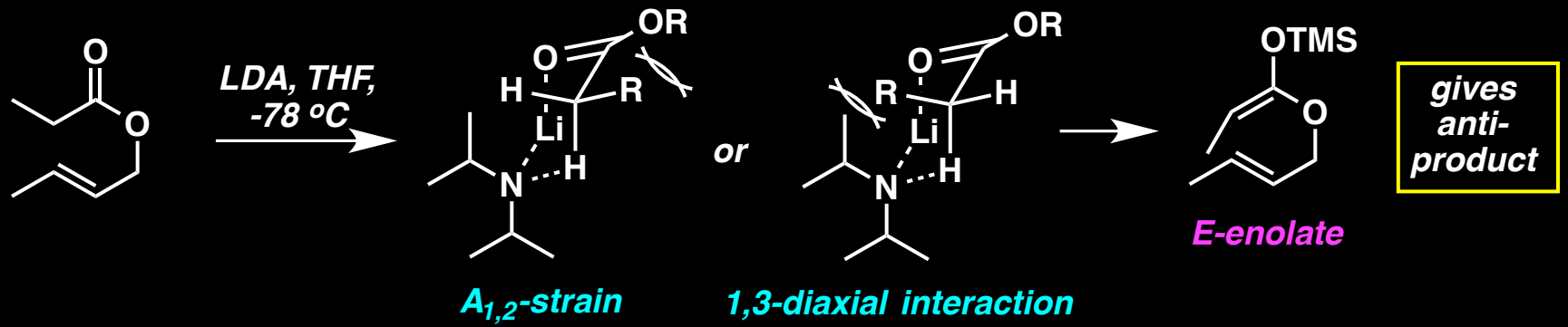


Key feature of chemistry is the ability to control enolate geometry to then control the formation of either syn- or anti- products

Work out the transition states for these on your own!

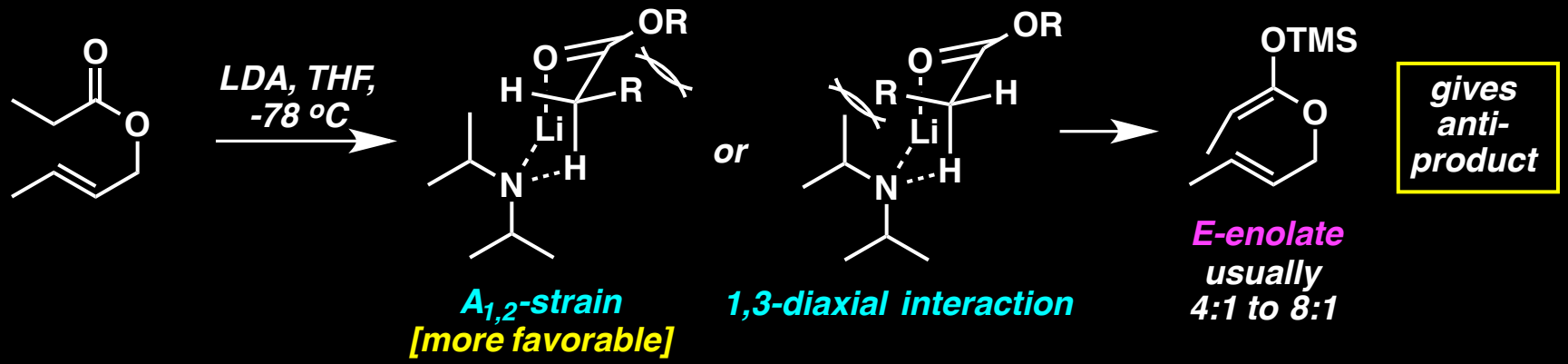
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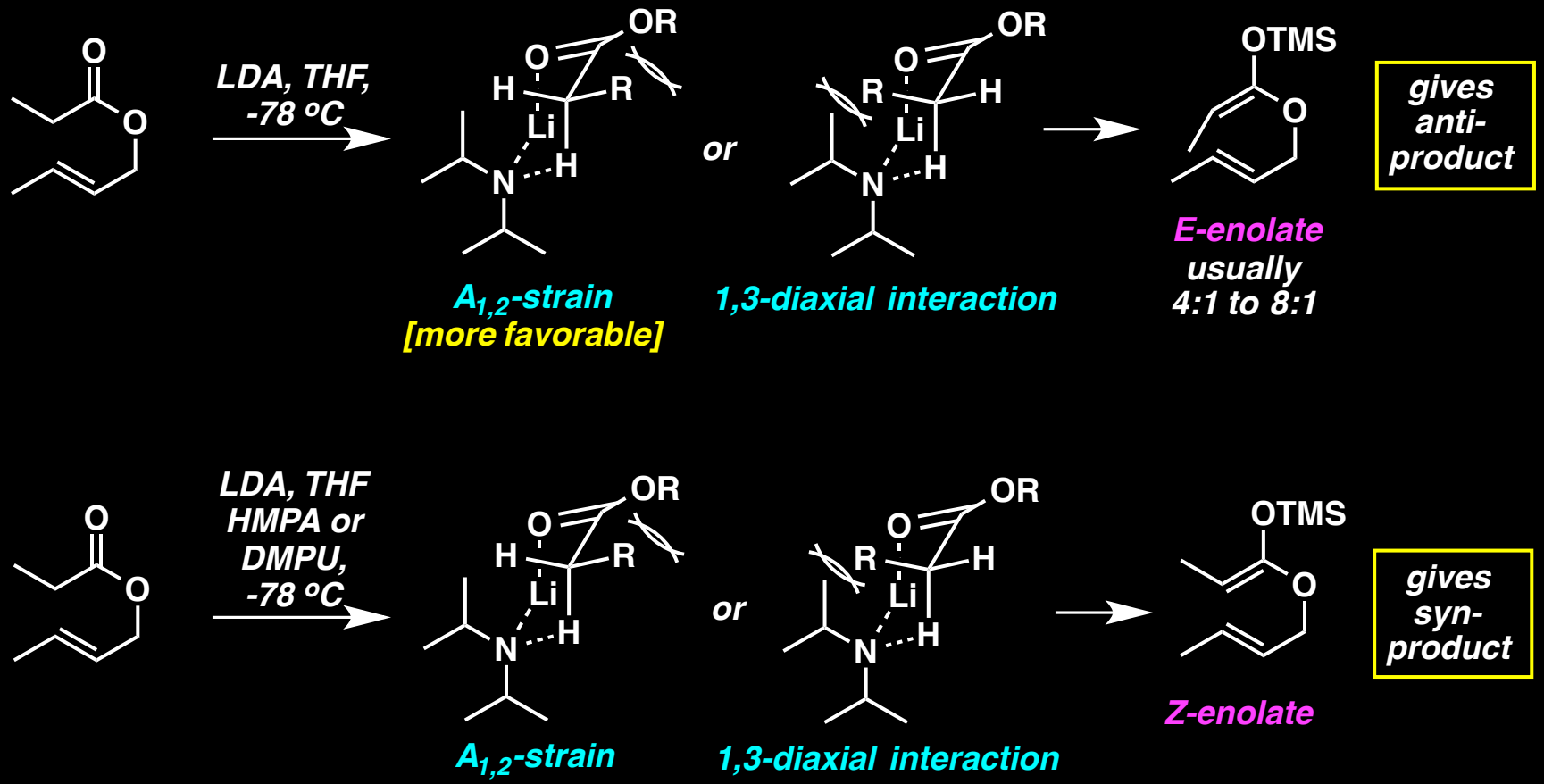
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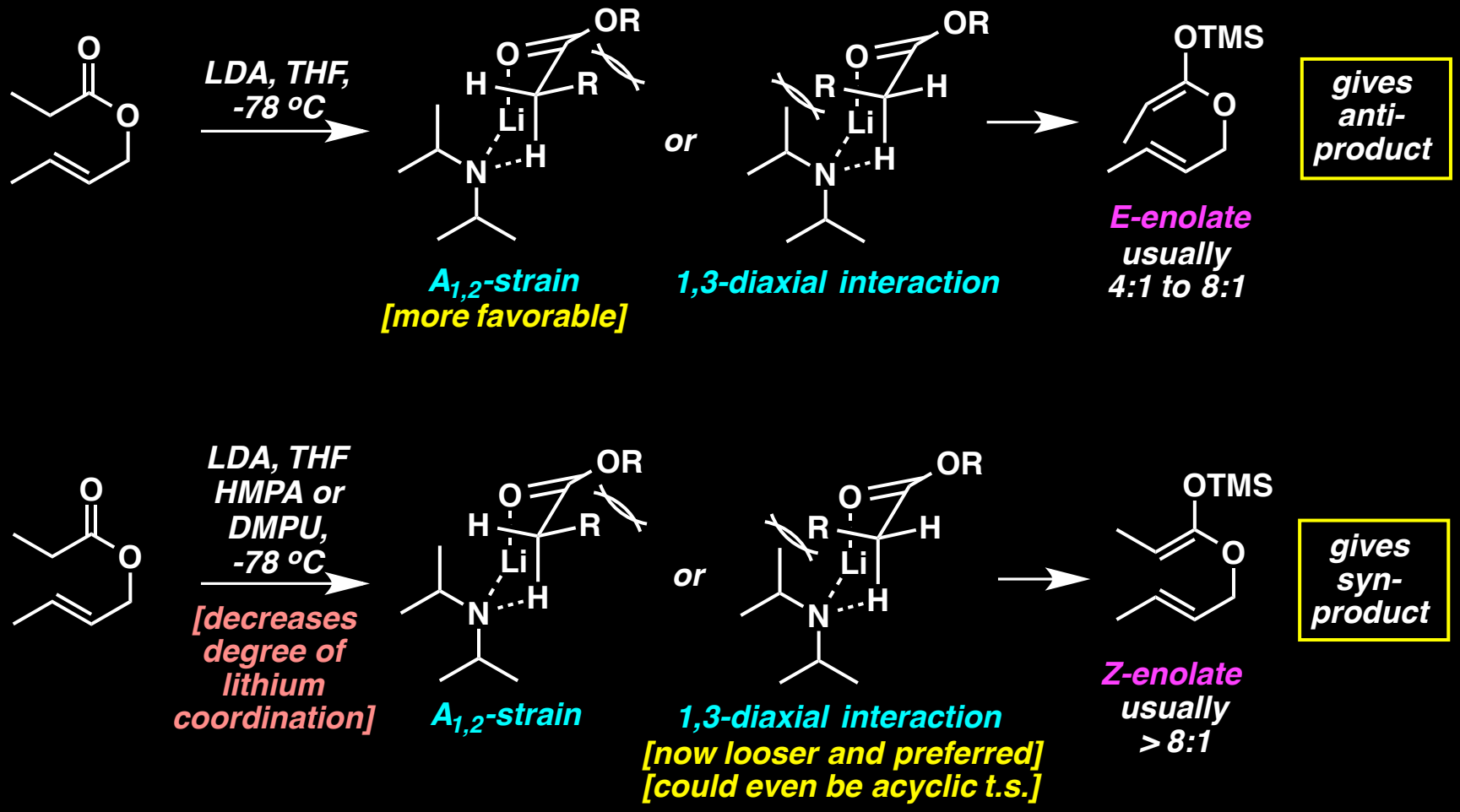
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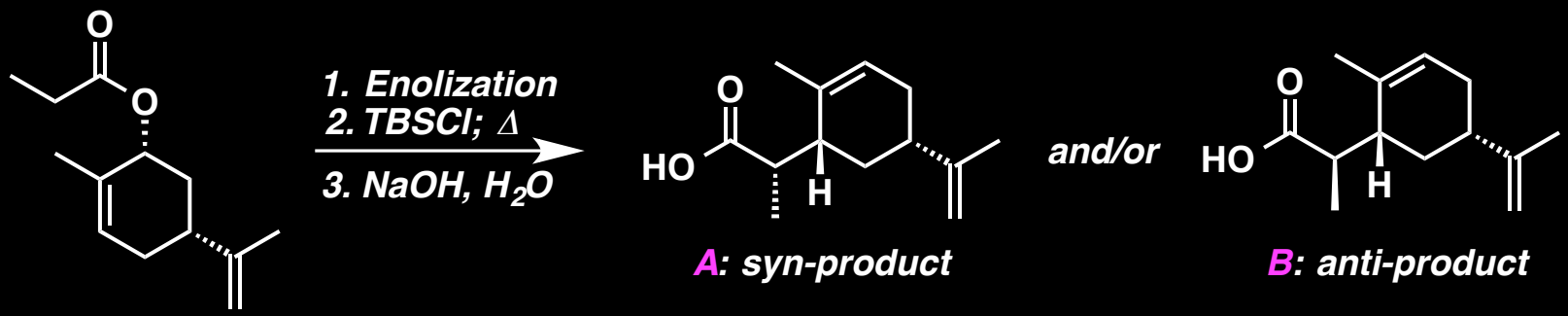
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The Ireland-Claisen Rearrangement: Use of Enolates to Generate Well-Defined Precursors



A: syn-product

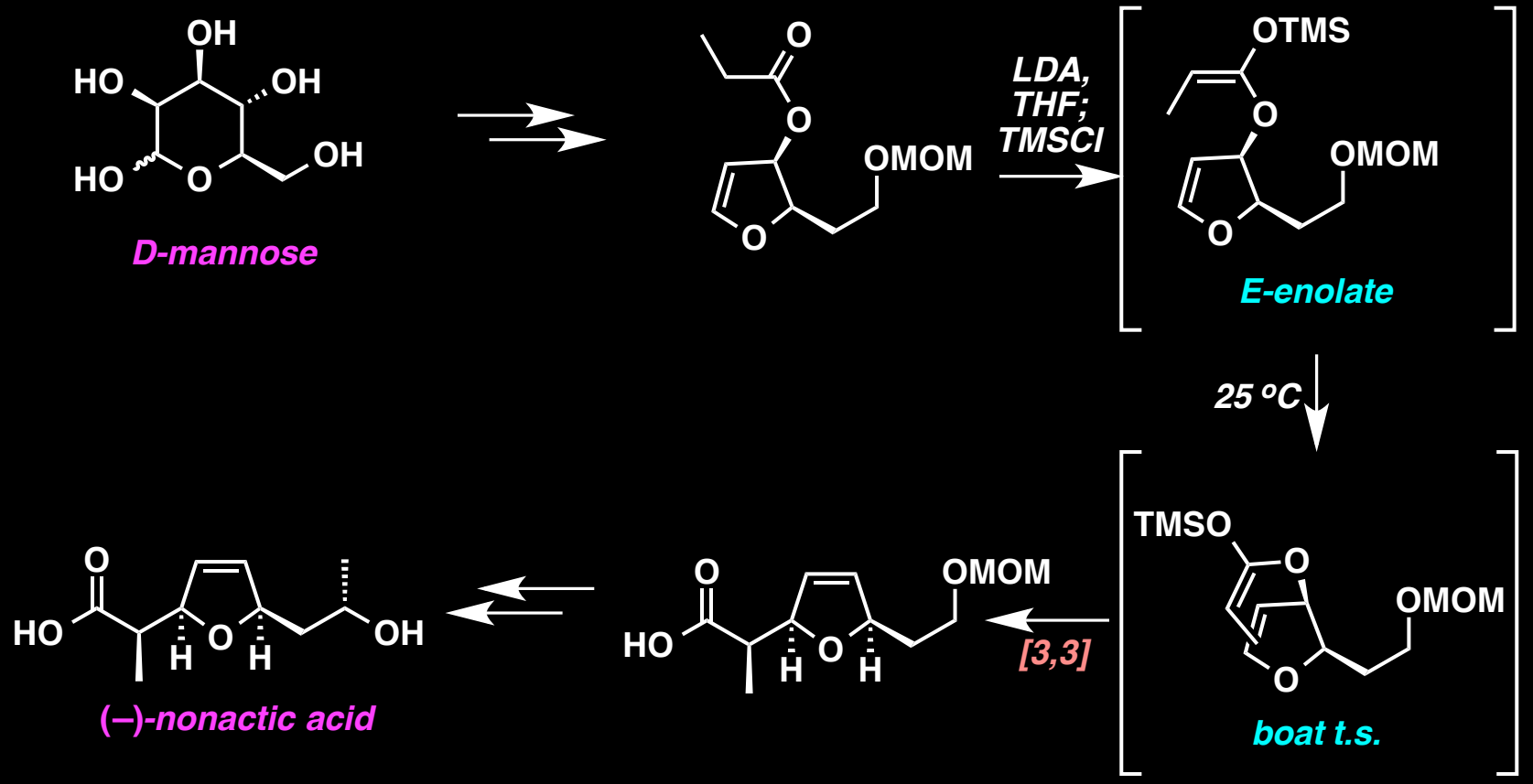
B: anti-product

Enolization conditions	A	B
LDA, 23% HMPA/THF	90	10
LDA, 45% DMPU/THF	98	2
LDA, THF	25	75

Remember: in cyclic systems boat-like transition states are possible. Here, the Z-enolate formed from the use of HMPA and DMPU likely goes through such a transition state. The standard chair should be favored with the formation of the E-enolate.

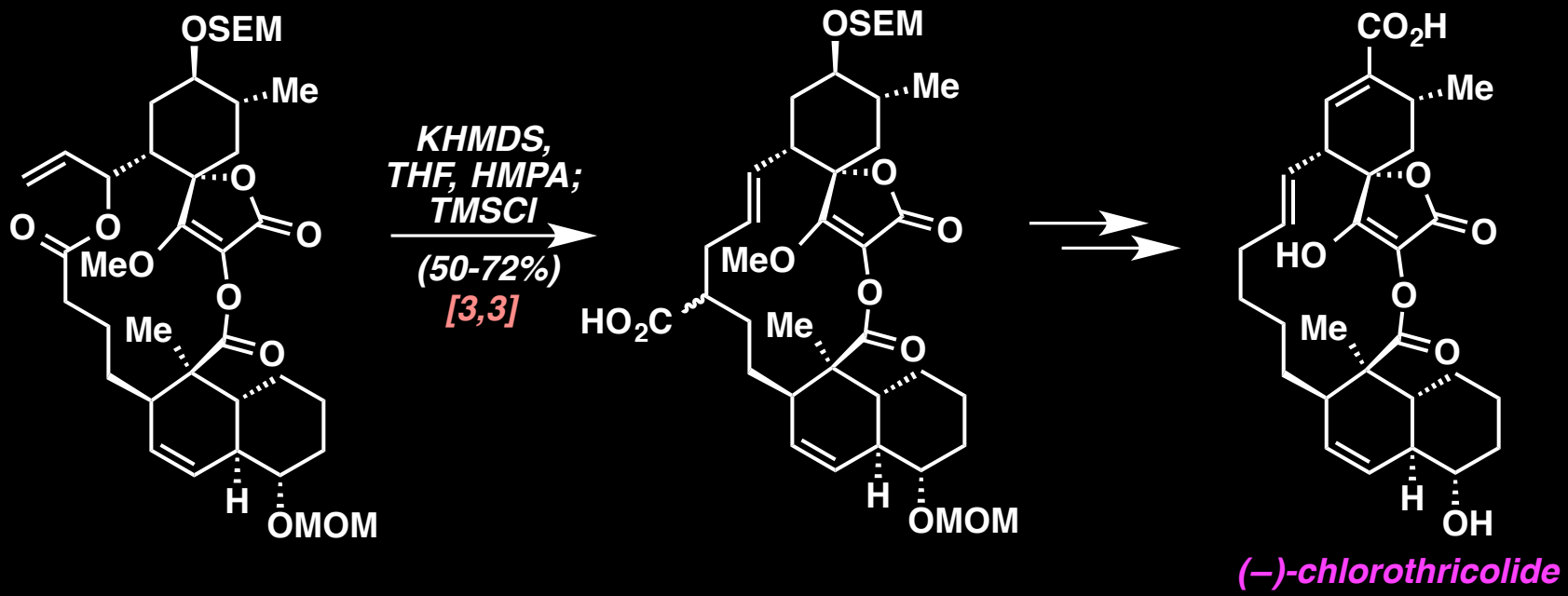
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The Ireland-Claisen Rearrangement: Applications in Complex Molecule Synthesis



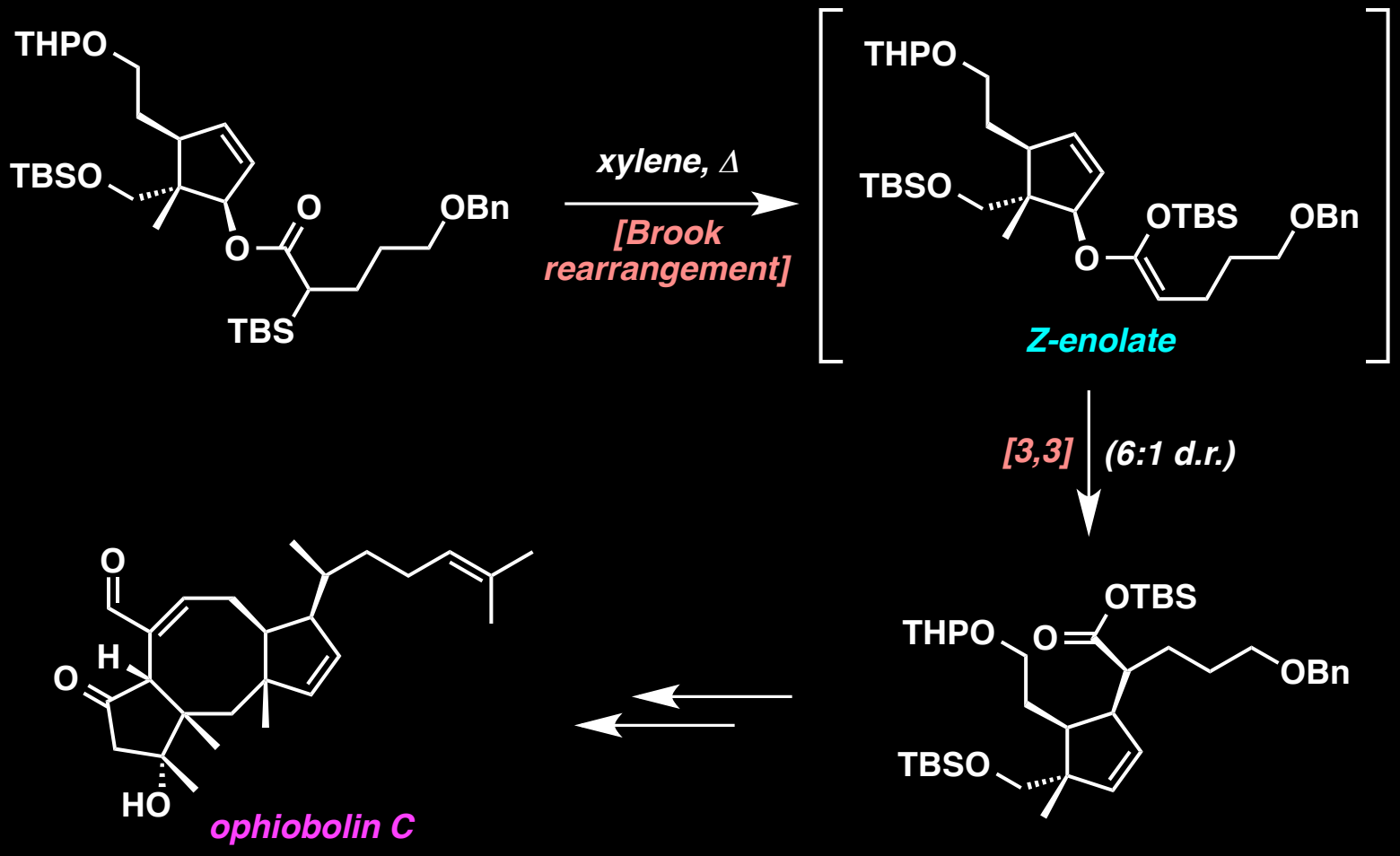
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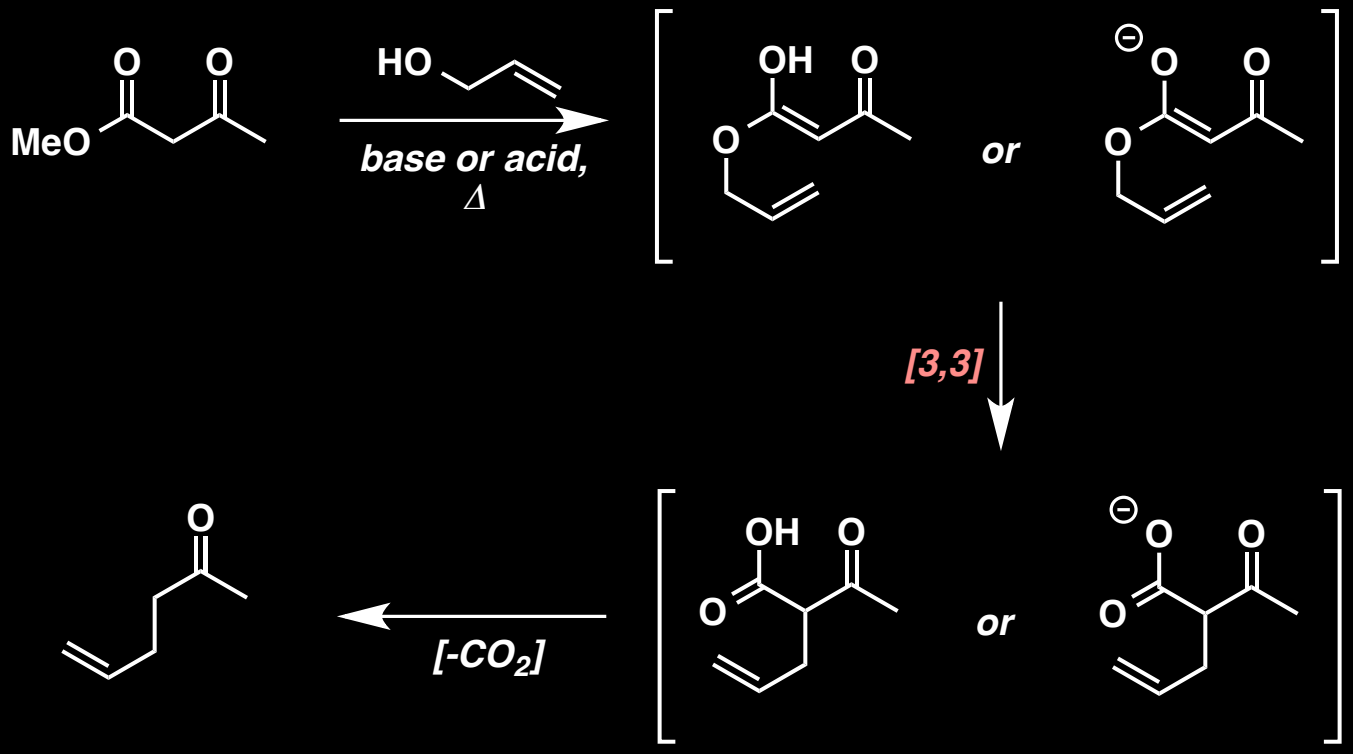
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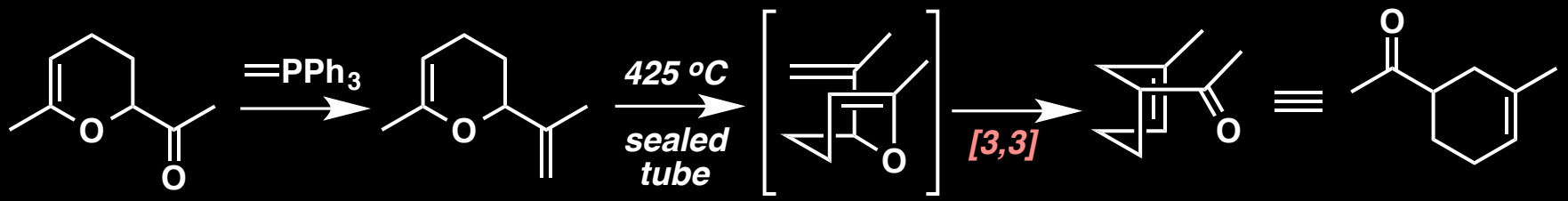
M. Rowley, M. Tsukamoto, Y. Kishi, J. Am. Chem. Soc. 1989, 111, 2735.

The Carroll Rearrangement: An Early Variant of the Claisen Reaction



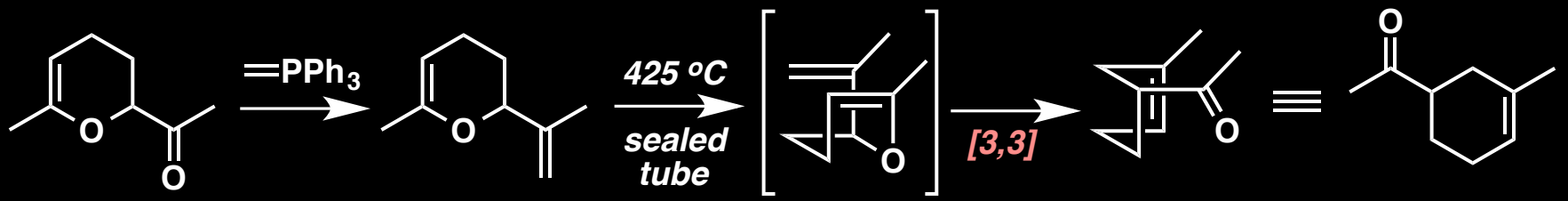
M. F. Carroll, J. Am. Chem. Soc. 1940, 62, 704.

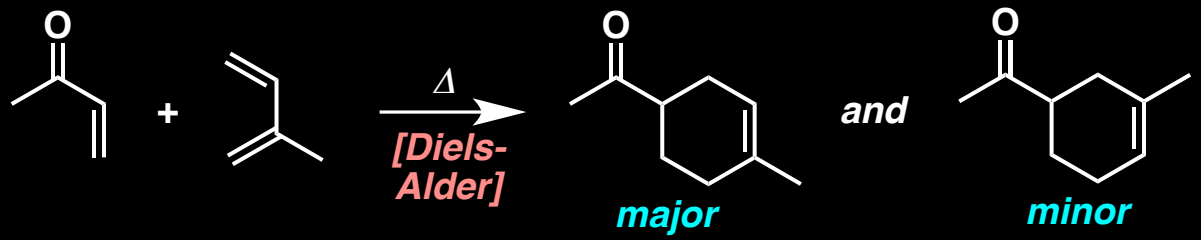
The Buchi-Claisen Rearrangement: A Method for the Preparation of Certain Cyclohexenes



G. Buchi, J. E. Powell, J. Am. Chem. Soc. 1970, 92, 3126.

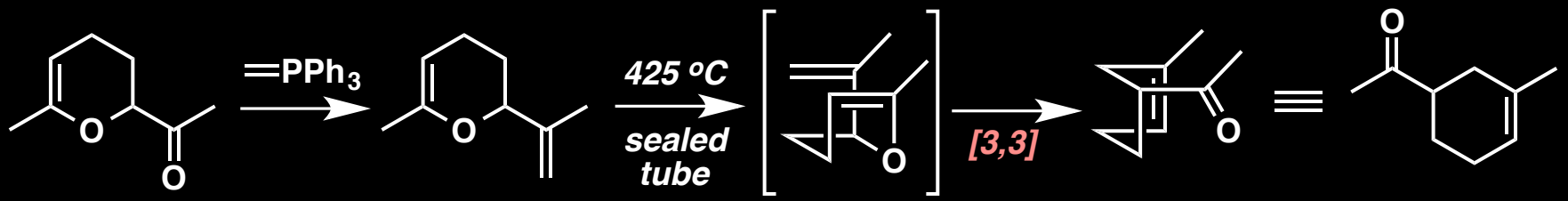
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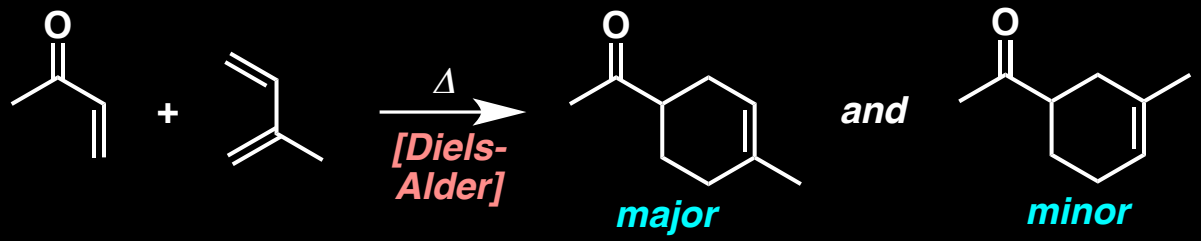




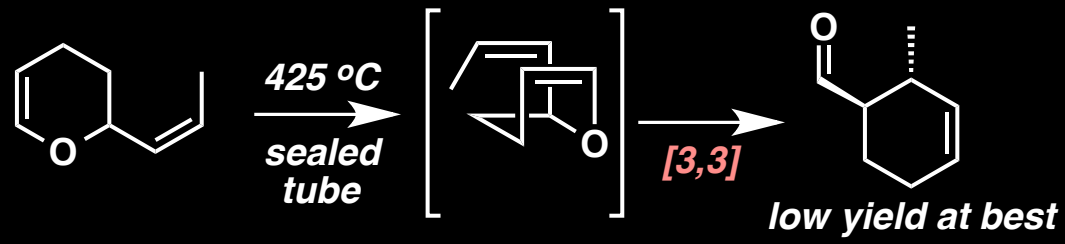
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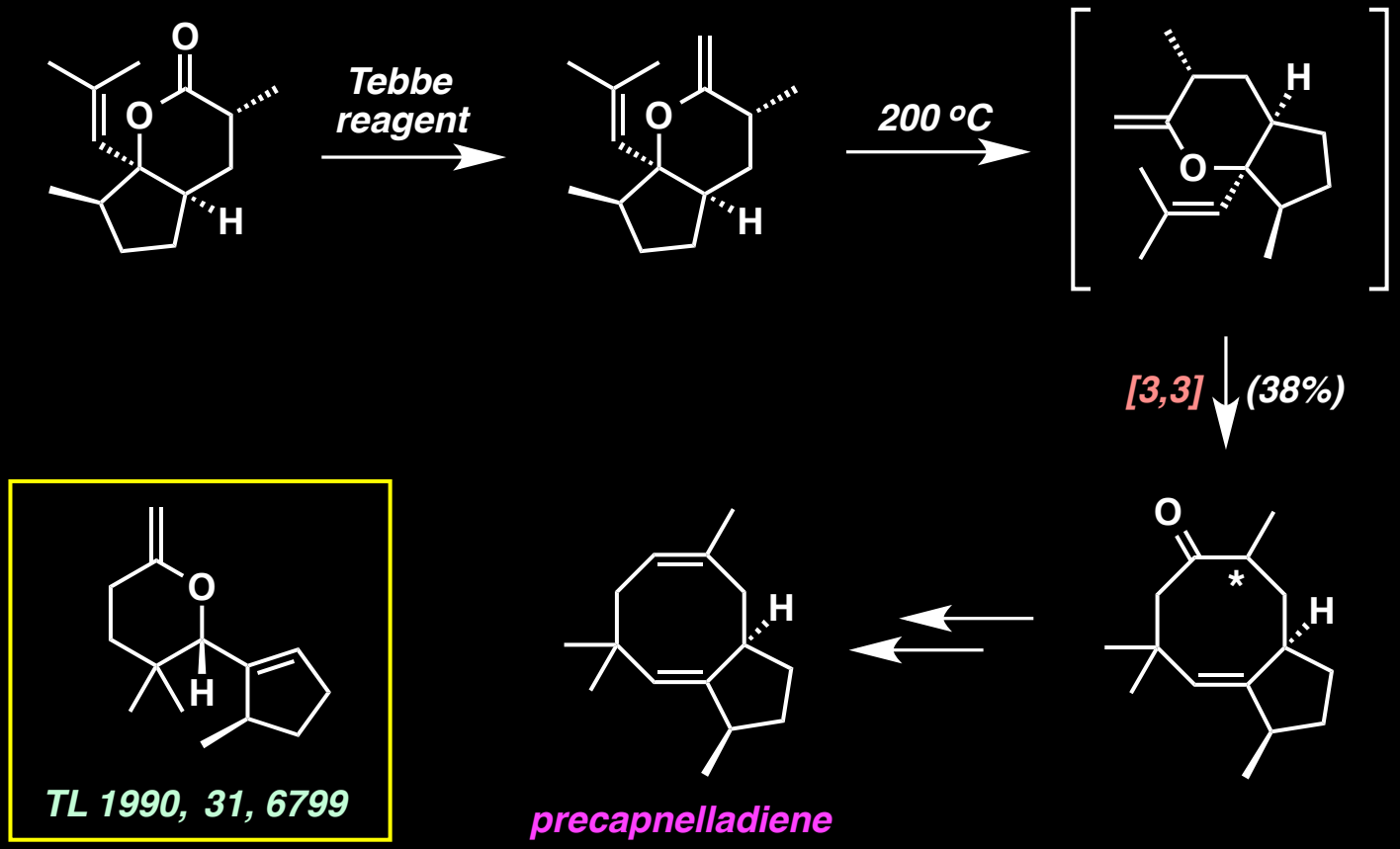


Reaction process must proceed via a boat transition state
Reaction temperatures are high
Cis-oriented exocyclic olefins afford poor yields of product



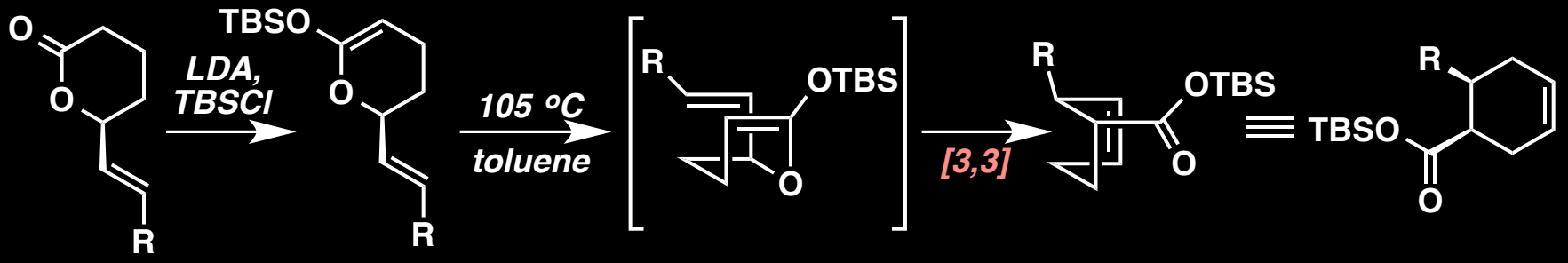
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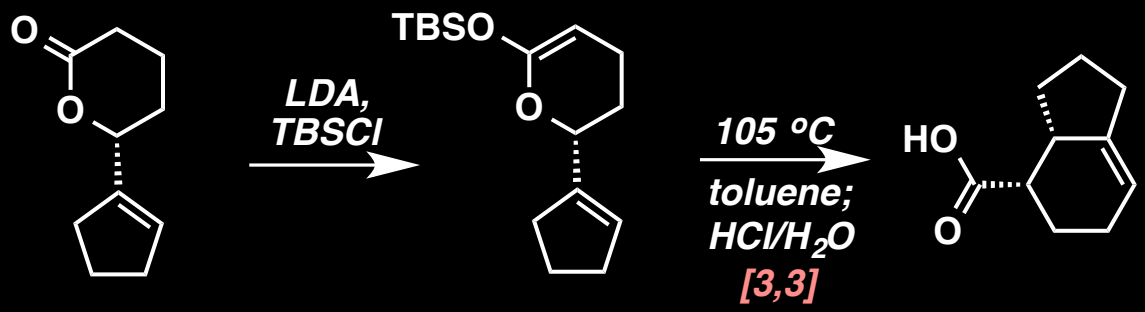
A Modified Buchi-Claisen Rearrangement: Large Ring Synthesis



W. A. Kinney, M. J. Coghlan, L. A. Paquette, J. Am. Chem. Soc. 1984, 106, 6868.

The Danishefsky-Claisen Rearrangement: Combining the Virtues of the Buchi- and Ireland-Claisen Reactions





S. J. Danishefsky, R. L. Funk, J. F. Kerwin, J. Am. Chem. Soc. 1980, 102, 6889.